Quantum Fields in de Sitter Spacetime

Simon Farache

Supervisor : Dionysios Anninos

September 2022

Abstract

The origin and future of the Universe are fundamental questions that theoretical physicists hold close to their hearts. The early inflation period and current acceleration of the Universe, described by de Sitter geometries stand at the heart of ongoing discussions. Here we review the fundamentals of such spacetimes and look at how basic quantum fields behave in such a background. We look at the significance of a vacuum state and show how the computation of a two-point function points in the direction of early de Sitter inflation. We also review some interesting concepts when working in de Sitter spacetime in ongoing research.

Contents

Introduction

Our Universe is accelerating, similarly to the earlier stages of its existence following the Big-Bang [1,28,39,40,41]. The way it is currently headed, the end of the stellifarous era will take place around 100 trillion years form now. This means the Universe will have expanded so much that the largest structures of the Universe, Galaxies, Clusters, will dilute into nothingness, and the only remaining source of energy will be that of the Cosmological Constant. Pure de Sitter spacetime is a specific solution to the Einstein Equations with positive Cosmological Constant, which approximates a distant future state of the Universe in which the only source of energy is the Cosmological Constant which is non dilute [1,26,34,39]. During it's initial expansion, inflation came to an end and the Universe entered a reheating phase in which the large structures formed. Under continued expansion, which is exponential in growth, we would have not been able access information as it would travel slower than this expansion, forming what we refer to as a cosmological horizon. Because early inflation came to an end, we have access to some information from the earliest stages that would have otherwise been beyond our cosmological horizon. This comes in the form of the Cosmological Microwave Background, primordial radiation dating from these early stages [1,39,42]. In a sense, we stand as a "meta-observer" of the early stages of the Universe, by accessing this data which is confined beyond the cosmological horizon of this early Universe.
Recent observations suggest that we are entering a similar phase, with a small non-vanishing Cosmological Constant [1,40,41]. Hence studying de Sitter spacetimes and quantum fields (observables) in this background could unveil some insights into the earlier stages of the Universe and the future of our Universe and maybe pave the way to answer fundamental questions from philosophical and theoretical physics points of view: Where do we come from and where are we going ? [39]
De Sitter spacetimes however present some challenges and conceptual difficulties when studying observables in this background. Indeed as we will see, most of the notions we are familiar with when studying quantum fields disappear due to the structure of de Sitter spacetime geometry. We will review the fundamental structure of de Sitter spacetimes and how those issues arise, before we can look at quantum fields and the two-point function of the massless scalar field in this background. We will finish by reviewing some interesting concepts from ongoing research.

1 de Sitter Spacetime

1.1 Asymptotically de Sitter Spacetime

Asymptotically de Sitter spacetimes can be considered as a class of spacetime geometries. These geometries, which contain a lot of extra features (compared to pure de Sitter), when viewed from a local region and asymptotically expanded in the future, approximate to pure de Sitter spacetime and lose those extra features [1,2]. The reason why this class is important is that pure de Sitter refers to a spacetime in the absence of matter and energy apart from the Cosmological Constant. If we have evidence showing that our Universe is accelerating [40,41], which can be described by de Sitter geometries in the Inflationary model [9,28], we know that it contains a non trivial amount of matter in comparison to the Cosmological Constant, which deforms the geometry of pure de Sitter, and therefore has a geometry asymptotically de Sitter. Although pure de Sitter allows for an interesting study of the features of this Universe, reality is closest represented by asymptotically de Sitter spactimes.

In this section we define this class of spacetime geometries as a solution of Einstein's equations and give an overview of its different geometrical features which allows us to have an insight into the spacetime's causal structure. We then study its isometries by looking at the Lie group and Lie algebra generators of the isometry group of de Sitter as a topological manifold. We will then look at some representation theory of the isometry group. We define (future) asymptotically de Sitter space time in the following way [2,4,6].

Definition 1.1. A spacetime $\{\mathcal{M},\, g_{ab}\}$ satisfying Einstein's equation with a positive cosmological constant

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 0, \tag{1}$$

$\Lambda > 0$, is asymptotically de Sitter if $\exists$ a manifold $\tilde{\mathcal{M}}$ with a boundary $\mathcal{I}^+$ (infinite future) and metric $\tilde{g}_{ab}$ and a diffeomorphism $\varphi : \tilde{\mathcal{M}} \longrightarrow \mathcal{M}$ such that :

- $\exists$ a smooth function Ω on $\tilde{\mathcal{M}}$ with :

- $\Omega = 0$ on $\mathcal{I}^+$,

- $\nabla_a \Omega$ is nowhere vanishing on $\mathcal{I}^+$,

- the unphysical metric $\tilde{g}_{ab}$ maps to the physical one via $\tilde{g}_{ab} = \Omega^2 g_{ab}$.

- The induced metric on $\mathcal{I}^+$ is locally isometric to the metric on the S^{d-1} sphere.

The length scale of de Sitter spacetimes is related to the Cosmological Constant by $\ell = \sqrt{\frac{(d-1)(d-2)}{2\Lambda}}$ [1,2,3,5]. A deeper analysis of Einstein equations is required to further characterise the asymptotically de Sitter metrics, which is beyond the scope of our discussion. Asymptotically de Sitter can be used for various applications, like the study of black holes where we take $\mathcal{I}^+$ with topology of $\mathbb{R} \times S^{d-2} \cong S^{d-1} \setminus \{p_1, p_2\}$ or Cosmology where we work with Asymptotically de Sitter in "Poincaré Coordinates" where $\mathcal{I}^+$ has topology of $\mathbb{R}^{d-1} \cong S^{d-1} \setminus \{p\}$ [2]. For our discussion on quantum fields in a de Sitter background, we only need pure de Sitter, as we are interested in late-time behaviour of observables.

1.2 Geometry of pure de Sitter space and Causal Structure

As stated in Definition 1, pure de Sitter (as a member of asymptotically de Sitter spacetime) solves equation (1), i.e $G_{\mu\nu} = 0$ with $\Lambda = \frac{3}{\ell^2}$[1,4,6]. The Cosmological Constant is valued at $\Lambda \sim 10^{-52} m^{-2}$ [1].

De Sitter space can be thought of as a four dimensional hyperboloid embedded in 5D Minkowski Space where Minkowski space has metric $ds^2 = -(dX^0)^2 + \sum_{i=1}^{4}(dX^i)^2$. Note the use of the convention $\eta = (-, +, +, +, +)$ which is used herafter.

The global geometry of de Sitter space is described by the metric [1,4]

$$ds^2 = -d\tau^2 + \ell^2 \cosh^2(\tau/\ell)(d\psi^2 + \sin^2\psi d\Omega_2^2), \tag{2}$$

with $\tau \in \mathbb{R}$, $d\Omega_2^2$ is the round metric on the two-sphere S^2. Note that here we use the convention $c(=\hbar) = 1$. Otherwise, there would be a c^2 factor before the timelike differential. This metric can also be written in terms of the three-sphere metric [2,3]

$$ds^2 = -d\tau^2 + \ell^2 \cosh^2(\tau/\ell) d\Omega_3^2, \tag{3}$$

where equation (2) is just the two-sphere "foliation" of this metric (the word foliation is put in quotation marks because a foliation is normally used when foliating a manifold with spacelike slices, but is used here for lack of a better word). [3]. We can notice that the presence of a τ-dependent function in the metric means the absence of a ∂_τ Killing vector on the manifold, generator of time-translation symmetry, which indicates the disappearance of energy conservation on the global de Sitter geometry, which, for the theoretical physicist, is rather concerning. We will come back to this in greater detail later. Constant τ sections of the geometry are 3-spheres which shrink from $\mathcal{I}^-$ at $\tau = -\infty$ to $\tau = 0$ and grow from $\tau = 0$ to $\tau = +\infty$ at $\mathcal{I}^+$ [1].

We can look at descriptions of different parts of the global geometry for various purposes, notably marking the difference between a meta-observer which has access to a finite region of the future boundary $\mathcal{I}^+$, and a static patch observer which only has access to a restricted view, confined to a thermal cavity [1]. Various descriptions are given [1,5] as follows.

Planar metric

It is sometime more convenient to use planar spacelike slices to describe half of the global geometry. Those planar slices are described by the metric

$$ds^2 = -d\tau^2 + e^{\frac{2\tau}{\ell}} d\vec{x}^2 = \frac{1}{\eta^2}\left(-\ell^2 d\eta^2 + d\vec{x}^2\right). \tag{4}$$

$\eta = -e^{-\tau/\ell}$ is known as the "conformal coordinate" with $-\infty < \eta < 0$. The planar metric describes the space inside the lightcone emanating from a point at $\mathcal{I}^-$ $(\tau \to -\infty)$. To access the region inside the lightcone emanating from $\mathcal{I}^+$, we simply perform the coordinate transformation $\tau \longrightarrow -\tau$, $\eta \longrightarrow -\eta$. Note the sign change of the time coordinate, which keeps the time "moving forward". Global and planar geometries can't be accessed fully be a single observer, only by what we referred to earlier as a meta observer, as parts of the geometry are separated by the light cone as is illustrated in Figure 1. $45°$ lines across the geometry are null geodesics and determine the lightcone of a single observer from the point at which they emanate. Constant time slices are hyperbolic curves reaching infinity tangentially where the light cone ends at $\mathcal{I}^+$. The causal structure of de Sitter showed by the Penrose diagrams shows clearly that some regions are inaccessible to a single observer as they are causally unrelated. To describe the space accessible to a single observer, we use what is known as the static patch.

Static Patch

This patch of de Sitter space is described by the metric

$$ds^2 = -\left(1 - \frac{r^2}{\ell^2}\right) dt^2 + \left(1 - \frac{r^2}{\ell^2}\right)^{-1} dr^2 + r^2 d\Omega_2^2, \tag{5}$$

with $r \in [0, \ell]$, $t \in \mathbb{R}$. Contrary to the global or planar geometry, the static patch does have ∂_t Killing vector, with a vanishing norm at $r = \ell$. $r = \ell$ is a null surface known as the cosmological horizon, which surrounds the observer at all times. This is the limit to the causally related fraction of de Sitter to which the single observer has access. The static patch is shown is Figure 1. The Future "triangle" as it is illustrated in the Penrose diagram, is described by the same metric by taking $r \in (\ell, \infty)$.

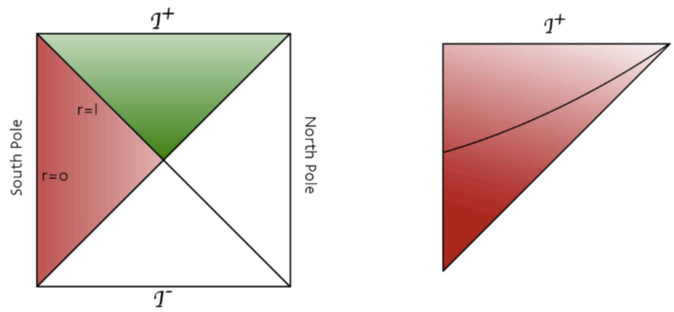

Figure 1: Penrose diagrams of de Sitter spacetime (taken from [1]).
Left: The full square is the global patch, where each point is a 2-sphere. The red triangle represents the Southern static patch and the green one is the future triangle.
Right: The triangle is the future directed planar patch, which contains $\mathcal{I}^+$ and the central line is a constant time $\mathbb{R}^3$ slice, tangent to the null geodesic.

We can also consider a geometry which includes the Static Patch and the Future triangle and is smooth across the horizon. It is described by the metric

$$\frac{ds^2}{\ell^2} = (-\frac{\rho}{\alpha} + \rho^2)d\tau^2 + d\tau d\rho + (1 - 2\rho\alpha)^2 d{\Omega_2}^2, \tag{6}$$

where $\tau \in \mathbb{R}$ and $\rho \in (1/2\alpha, -\infty)$. Note that $\mathcal{I}^+$ is located at $\rho = -\infty$ and constant ρ slices are null rays emanating from the worldline at $\rho = 1/2\alpha$ and ending at $\mathcal{I}^+$. The horizon of this geometry is located at $\rho = 0$.

Other geometries

Other patches are useful for convenience purposes in different studies than what is to be pursued here, so we will give a few briefly for completeness. The Hyperbolic patch of de Sitter is given [1] as described by the metric

$$ds^2 = -d\tilde{t}^2 + \ell^2 \sinh^2\left(\frac{\tilde{t}}{\ell}\right) d\mathcal{H}_3^2, \tag{7}$$

where $\tilde{t} \in [0, \infty)$ and $d\mathcal{H}_2^2 = dR^2 + \sinh^2 R d\Omega_2^2$ is the metric of hyperbolic three-space.
The de Sitter/de Sitter patch is dS_4 (four dimensional de Sitter space) radially foliated by

three-dimensional de Sitter space. It's metric is given [1] as

$$ds^2 = dw^2 + sin^2\left(-d\tilde{\tau}^2 + \ell^2\cosh^2(\tilde{\tau}/\ell)d\Omega_2^2\right), \tag{8}$$

with $w \in [0, \pi]$, $\tau \in \mathbb{R}$.
Another patch is de Sitter foliated by $\mathcal{H}_2 \times S^1$ slices

$$ds^2 = -\frac{d\tau^2}{(1 + (\tau/\ell)^2)} + \left(1 + (\tau/\ell)^2\right)dx^2 + \tau d\mathcal{H}_2^2, \tag{9}$$

with $\tau \in [0, \infty)$, $x \sim x + \ell$, $d\mathcal{H}_2^2$ is the hyperbolic metric of two-space.

In the case of pure de Sitter, the isometry group manifests itself as the conformal group of the three-metric on $\mathcal{I}^{\pm}$ which can be $\mathbb{R}^3$, $S^2 \times \mathbb{R}$, $\mathcal{H}_3$, $\mathcal{H}_2 \times S^1$, or S^3 [1]. We now go into details through these isometries of de Sitter space and the Killing vectors associated with them.

1.3 Isometries and Killing Vectors

1.3.1 Mathematical background

When working on a Riemannian manifold, or pseudo-Riemannian in the case of de Sitter spacetime, it is important to consider the preservation of its metric structure. This is why we study isometries.

Definition 1.2. An isometry between two (pseudo-)Riemannian manifolds $\mathcal{M}$ and $\mathcal{N}$ is a diffeomorphism $\varphi\colon \mathcal{M} \to \mathcal{N}$ such that, $\forall\, v, w \in T_p\mathcal{M}$ (tangent space of $\mathcal{M}$ at point p)

$$g_p^{\mathcal{M}}(v, w) = g_{\varphi(p)}^{\mathcal{N}}\left(d\varphi_p(v), d\varphi_p(w)\right) \tag{10}$$

where v and w are tangent vectors, $g_p^{\mathcal{M}}(v, w)$ is the metric on $\mathcal{M}$ at p and $g_{\varphi(p)}^{\mathcal{N}}\left(d\varphi_p(v), d\varphi_p(w)\right)$ is the metric on $\mathcal{N}$ at $\varphi(p)$ and $d\varphi_p$ is the differential of φ at p acting on a vector field (also known as the push forward).
An isometry of $(\mathcal{M}, g)$ is a diffeomorphism $\varphi\colon \mathcal{M} \to \mathcal{M}$.

In other words, an isometry transformation on the manifold is one which leaves the metric unchanged, i.e the metric structure (distances) are preserved.
The set of isometries on a manifold forms its isometry group, with an associated Lie algebra, which is isomorphic to the tangent space of the manifold at the identity $T_I\mathcal{M}$. The Killing

vectors, are representations of the infinitesimal generators of the isometry group algebra in the tangent space [4].

Now we know that isometries preserve the metric structure on the manifold, and Killing vector fields can be viewed as infinitesimal generators of these transformations, we need to define Killing vectors on the manifold.

Formally, a vector field is called a Killing vector field if its flow Φ_t is a local isometry of $(\mathcal{M}, g)$, $\mathcal{M}$ the manifold, g its local metric. A useful tool in studying flows on a manifold is the Lie derivative which, without going into unnecessary detail, expresses the leading order change in the metric under an infinitesimal transformation $\delta\xi$ along its flow (when acting on the metric with respect to a vector field along its flow) [4]. This can be expressed as

$$\mathcal{L}_{\delta\xi} g_{\mu\nu} = \delta\xi^{\sigma}\partial_{\sigma} g_{\mu\nu} + g_{\mu\sigma}\partial_{\nu}\delta\xi^{\sigma} + g_{\sigma\nu}\partial_{\mu}\delta\xi^{\sigma}. \tag{11}$$

It is in turn easy to show, using the definition of the covariant derivative in terms of Christoffel symbols, that the Lie derivative can in turn be expressed in terms of covariant derivatives in the tangent space as

$$\mathcal{L}_{\delta\xi} g_{\mu\nu} = \nabla_{\mu}\delta\xi_{\nu} + \nabla_{\nu}\delta\xi_{\mu}. \tag{12}$$

From Definition 1.2 together with the Lie derivative, we can define a Killing vector as a vector field $\delta\xi^{\mu}$ such that

$$\mathcal{L}_{\delta\xi} g_{\mu\nu} \equiv \nabla_{\mu}\delta\xi_{\nu} + \nabla_{\nu}\delta\xi_{\mu} = 0. \tag{13}$$

Vectors which satisfy the Killing equation are representations of the infinitesimal generators of isometry group of the manifold and the Killing vectors furnish its Lie algebra and close under Lie brackets. We can introduce a variant to Killing equation, through conformal transformations, under which the metric remains proportional to the original one up to some factor known as the conformal weight. The equation is known as the conformal Killing equation, and the vectors that satisfy this equation are representations of infinitesimal generators of the Conformal group, and furnish its Lie algebra [7].

This gives us context into studying the isometries of de Sitter spacetime and why they are important when studying quantum fields in curved spacetime.

1.3.2 Isometries from the embedding

Solving the Killing equation directly in de Sitter space can be very long and analytically tiedious. A more efficient way of obtaining them is to consider dS_4 in its ambient space.

As mentioned earlier, four-dimensional de Sitter can be viewed as a hyperboloid embedded in 5D Minkowski spacetime $\mathbb{R}^{1,4}$. The isometries are inherited from the ambient space by restricting them to the hypersurface. This hypersurface is defined as

$$\Sigma: \; -(X^0)^2 + \sum_{i=1}^{4}(X^i)^2 = \ell^2. \tag{14}$$

In other words, the isometries of dS_4 are the isometries of $\mathbb{R}^{1,4}$ which preserve the hyperboloid, namely the isometries associated with the Killing vector fields [3]

$$\begin{aligned} \delta\xi^{\mu}_{0i}\partial_\mu &= X^0\partial_i + X^i\partial_0 \\ \delta\xi^{\mu}_{ij}\partial_\mu &= X^0\partial_\alpha - X^\alpha\partial_0 \qquad i,j = 1,2,3,4. \end{aligned} \tag{15}$$

More specifically, the isometry group is the set of transformations that leave
$x^2 + y^2 + z^2 + w^2 - t^2 = \ell^2$ invariant, which is precisely SO(1,4). Its algebra is $\mathfrak{so}(1,4)$ and the Killing vectors are representations of the generators. This group is 10 dimensional, it has $\frac{n(n-1)}{2} = 10$ degrees of freedom, and thus we expect to have 10 Killing vectors. We will discuss the Lie algebra further in Section 1.3.3
The embedding coordinates on de Sitter can be taken as

$$\begin{aligned} X^0 &= \ell\sinh(\frac{\tau}{\ell}) \\ X^i &= \ell\cosh(\frac{\tau}{\ell})z_i, \end{aligned} \tag{16}$$

where $\sum_{i=1}^{4} z_i^2 = 1$ the unit sphere in $\mathbb{R}^4$. The z_i are given as $z_1 = \sin(\psi)\sin(\phi)\cos(\theta)$, $z_2 = \sin(\psi)\sin(\phi)\sin(\theta)$, $z_3 = \sin(\psi)\cos(\phi)$, $z_4 = \cos(\psi)$. By restricting the differentials of the X^i to the hypersurface (Eq.14)

$$\begin{aligned} dX^0|_\Sigma &= \cosh\left(\frac{\tau}{\ell}\right)d\tau \\ dX^i|_\Sigma &= \sinh\left(\frac{\tau}{\ell}\right)d\tau + \ell\cosh\left(\frac{\tau}{\ell}\right)dz_i\,, \end{aligned} \tag{17}$$

it is quite straightforward to show that we recover the metric of de Sitter spacetime (Eq.2). However, evaluating the Killing vectors on the manifold is far less trivial. The direct restriction of partial derivatives is analytically long and complex. As a way around this, we can evaluate the Killing vectors by constructing a scalar quantity on the manifold. The simplest object to restrict to Σ is a scalar field $\Phi(X^A)$ and is evaluated as $\Phi(X^A)|_\Sigma = \Phi(X^A|_\Sigma)$. We build the scalar quantity out of the Killing vector by contracting with the differential dX^A and evaluating it on Σ

$$g^{ind}_{\mu\nu}\delta\xi^{\mu}_{\alpha\beta}dx^\nu = \eta_{AB}\delta\xi^{A}_{\alpha\beta}\,dX^B\big|_\Sigma, \tag{18}$$

where η is the metric on the ambient Minkowski spacetime.

For simplicity's sake we do it here for two-dimensional de Sitter and extrapolate the isometry generators to four-dimensional de Sitter. The isometry group of two-dimensional de Sitter is SO(1,2), so we have 3 Killing vectors (3 degrees of freedom of SO(1,2)). We simply replace the ambient 5-D Minkowski space with 3D-Minkowski, where the z_i are now the unit circle, i.e $\sum_{i=1}^{2} z_i^2 = 1$ and follow the procedure outlined above. We briefly outline this procedure here, the full derivation can be found in Appendix A.

First, we define $z_1 = \sin\theta$, $z_2 = \cos\theta$, and the differentials $dX^0|_\Sigma = \cosh(\frac{\tau}{\ell})d\tau$, $dX^1|_\Sigma = \sinh(\frac{\tau}{\ell})\sin(\theta)d\tau + \ell\cosh(\frac{\tau}{\ell})\cos(\theta)d\theta$, $dX^2|_\Sigma = \sinh(\frac{\tau}{\ell})\cos(\theta)d\tau - \ell\cosh(\frac{\tau}{\ell})\sin(\theta)d\theta$. We use these along with equation (15) to evaluate equation (18) and the X^A values as given by equation (16), from which we get the Killing vectors as

$$\begin{aligned}
\delta\xi^\mu_{12}\partial_\mu &= -\partial_\theta \\
\delta\xi^\mu_{01}\partial_\mu &= \ell\sin(\theta)\partial_\tau + \tanh(\frac{\tau}{\ell})\cos(\theta)\partial_\theta \\
\delta\xi^\mu_{02}\partial_\mu &= \ell\cos(\theta)\partial_\tau - \tanh(\frac{\tau}{\ell})\sin(\theta)\partial_\theta\,.
\end{aligned} \tag{19}$$

Those Killing vectors are the generators of the isometry group O(1,2) on de Sitter, namely the infinitesimal generators of two boosts and one rotation. Extrapolating to dS_4, which has the isometry group SO(1,4), upon adding 2 dimensions of space, we have 4 boosts and 6 rotations. As mentioned earlier in Section 1.2, we notice the absence of a time-translation isometry Killing vector, and the absence of energy conservation on de Sitter space. Worrying indeed. We will deal with this issue when we start discussing fields, however one way around it is studying energy conservation the static patch, which does have a time-translation isometry Killing vector.

We now move on to the structure of its Lie algebra, which will prove essential further on, as we want to build a Hilbert space which behaves nicely under isometries.

1.3.3 Lie algebra of de Sitter

The Lie algebra matrices are the generators of the Isometry group SO(1,4). They are given [4] by

$$M(1,4) = \begin{pmatrix} 0 & b_1 & b_2 & b_3 & b_4 \\ b_1 & 0 & j_1 & j_2 & j_3 \\ b_2 & -j_1 & 0 & -j_4 & j_5 \\ b_3 & -j_2 & j_4 & 0 & -j_6 \\ b_4 & -j_3 & -j_5 & j_6 & 0 \end{pmatrix}, \tag{20}$$

where all the entries are real numbers. From these, via the one parameter group theorem of the Lie algebra of a linear Lie group, we can generate the Isometry group as
$\mathrm{SO}(1,4) = \{\exp(tM(1,4)) | t \in \mathbb{R}\}$ [11].
The matrix $M(1,4)$ is itself constructed of generators that furnish the Lie algebra $\mathfrak{so}(1,4)$ (not to be mistaken with the generators of SO(1,4)). SO(1,4) is 10 dimensional as it has 10 degrees of freedom, and its Lie algebra has 10 generators, each one associated with an infinitesimal generator of an isometry and its Killing vector on de Sitter space. Those generators are namely, as mentioned earlier, 4 boosts and 6 rotations, given [3,4] as

$$\begin{aligned} B_i \equiv B_1 = i \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, B_2 = i \begin{pmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \ldots \ , i = 1,2,3,4 \\ J_j \equiv J_1 = i \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, J_2 = i \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \ldots j = 1,2,3,4,5,6 \end{aligned} \tag{21}$$

Note the introduction of a factor i in the generators. This makes the generators Hermitian, which allows to match the form in which the commutation relations are computed, which is similar to that of the Poincaré group and of the conformal algebra of CFT's [17]. The reason why we note this similarity is because SO(1,4) is also the 3D Euclidean Conformal Group, and the generators can be rearranged such as to follow the Euclidean conformal algebra [23], ie. we define the symmetry generators of SO(1,4) as

$$J_{AB} = X_A \partial_B - X_B \partial_A \ ; \ \ A, B = 0,1,2,3,4 \tag{22}$$

and we define the conformal algebra generators

$$\begin{aligned} D = J_{04} \ , \quad M_{\mu\nu} = J_{\mu\nu}, \quad \mu, \nu = 1,2,3 \ \ \mu \neq \nu \\ P_\mu = J_{0\mu} + J_{4\mu} \ , \quad K_\mu = J_{4\mu} - J_{0\mu}. \end{aligned} \tag{23}$$

It follows from these definitions that the new generators follow the usual conformal algbera commutation relations. Note that these generators are anti-Hermitian and not Hermitian. This will prove useful when looking at the representation theory of the de Sitter group and when building a Hilbert space for our Quantum Field Theory.
We now come back to our original generators of the Lie algebra.

The procedure we follow here to compute the Lie brackets of the Lie algebra is described in full in [4].

We first look at the rotations generators commutation relations. We rename the variables as $J_i \equiv P_i$ for $i = 1, 2, 3$, $J_4 \equiv Q_3$, $J_5 \equiv Q_2$, $J_6 \equiv Q_1$. The commutation relations are given as

$$\begin{aligned}
[P_i, P_j] &= -i\varepsilon^{ijk} Q_k \\
[Q_i, Q_j] &= i\varepsilon^{ijk} Q_k \\
[P_i, Q_j] &= i\varepsilon^{ijk} P_k.
\end{aligned} \tag{24}$$

Now for the boosts Lie brackets. Let $B_i = K_{i-1}$ for $i = 1, 2, 3, 4$. The commutation relations are given as

$$\begin{aligned}
[K_0, K_i] &= -iP_i, \quad i, j = 1, 2, 3 \\
[K_i, K_j] &= -i\varepsilon^{ijk} Q_k, \quad i, j = 1, 2, 3.
\end{aligned} \tag{25}$$

The rest of the Lie brackets are given as

$$\begin{aligned}
[K_0, P_i] &= iK_i \\
[K_0, Q_i] &= 0 \\
[K_i, P_j] &= -i\delta_{ij} K_0 \\
[K_i, Q_j] &= i\varepsilon^{ijk} K_k
\end{aligned} \tag{26}$$

We thus see that the $\mathfrak{so}(1, 4)$ Lie algebra closes under Lie brackets on the manifold, which shows that the Killing vectors close under Lie brackets and therefore furnish the whole de Sitter algebra.

Before moving on, it is essential to stress that the notion associated with symmetries and thus isometries in a theory of gravity is not as trivial as we just described. All of the isometries we derived apply only to pure de Sitter space. Indeed, in a more realistic situation, the presence of matter would deform the geometry of the spacetime as we pointed earlier. Hence for Asymptotically de Sitter spacetimes, these isometries are destroyed [1]. It is essential when discussing fields to understand the relevance of these symmetries, especially with regards to the de Sitter invariant vacuum state, also known as the Bunch Davies state, which we will explore in greater detail later on. Before moving on to observables and Quantum Field Theory (QFT) in de Sitter spacetime, we look at unitary irreducible representation of the de Sitter algebra.

1.4 Unitary Irreducible Representations

As we will see later on, the notion of observables in de Sitter space is questionable. However, as we will need to build a Hilbert space for our QFT, we want to look at quantities that nonetheless behave nicely under the de Sitter group. This Hilbert space is organised in unitary irreducible representations of the de Sitter group in $\mathrm{SO}(1, d+1)$[23]. We come back to our dS_2 geometry (see §1.3.2), with the isometry group $\mathrm{SO}(1,2) \cong \mathrm{SL}(2,\mathbb{R})$ [23,24] for simplicity.

We can build infinite-dimensional representations of the isometry group labeled by a dimension λ. We are only interested in the trivial representation in this paper (which corresponds to a spin zero 0 field) so we will not look at the spin dimension of the representation [23].

If we have the three generators of our de Sitter algebra as {H,K,D}, with the commutation relations given by [26]

$$\begin{aligned}
&[\mathrm{D},\mathrm{H}]{=}\mathrm{iH}\\
&[\mathrm{K},\mathrm{H}] = 2i\mathrm{D} \\
&[\mathrm{D},\mathrm{K}] = -i\mathrm{K},
\end{aligned} \tag{27}$$

we can express new generators in terms of the original ones as

$$L_0 = \frac{\mathrm{H}+\mathrm{K}}{2} \quad ; \quad L_\pm = \frac{1}{2}\left(\mathrm{H}-\mathrm{K}\right) \mp i\mathrm{D} \tag{28}$$

with

$$[L_+, L_-] = 2L_0 \quad ; \quad [L_\pm, L_0] = \pm L_\pm. \tag{29}$$

L_0 is the generator of the SO(2) subgroup and $L_\pm$ are raising and lowering operators [26]. Note that the generators are hermitian, from the unitary condition and $L_- = (L_+)^\dagger$.

We can express the Casimir operator, respectively in terms of the new generators, as [24,26]

$$C = -L_0^2 + \frac{1}{2}(L_-L_+ + L_+L_-) \tag{30}$$

with eigenvalue $\lambda(1-\lambda)$ [24].

The representation is unitary for specific values of λ, which are divided in multiple categories, the three main ones given as [23,24,26,29]:

- $\lambda = \frac{1}{2} + i\nu$ with $\nu \in \mathbb{R}$ which consitutes the Principal Series irreducible representation,
- $\lambda \in (0, 1)$ which constitutes the Complementary Series irreducible representation.
- In odd dimensions of d, there is an additional series, the Discrete Series with $\lambda = n$ or

$\lambda = \frac{1}{2} + n,\ n \in \mathbb{N}$.

If we take basis vectors as $|\lambda, m\rangle$ with $m \in \mathbb{Z}$, we have [24]

$$\begin{aligned} &L_0|k,m\rangle = -m|k,m\rangle, \\ &\langle\lambda, m|L_+L_-|\lambda, m\rangle = (m-\lambda)(m-1+\lambda), \\ &\langle\lambda, m|L_-L_+|\lambda, m\rangle = (m+\lambda)(m+1-\lambda). \end{aligned} \tag{31}$$

In addition to the operators we introduced above, we can build al class of operators $\Theta_m^{(\lambda)}$ and $(\Theta_m^{(\lambda)})^\dagger$ that transform nicely under the de Sitter group [26]. These are known as conformal operators and, focusing on the principal series, they are defined as

$$\begin{aligned} &[L_0, \Theta_m^{(\lambda)}] = m\Theta_m^{(\lambda)} \\ &[L_\pm, \Theta_m^{(\lambda)}] = -(m \pm \lambda)\Theta_{m\pm1}^{(\lambda)}. \end{aligned} \tag{32}$$

Given a de Sitter invariant vacuum state $|\Omega\rangle$, we can define a single particle state at $\mathcal{I}^+$

$$\Theta_m^{(\lambda)}|\Omega\rangle \equiv |\lambda, m\rangle. \tag{33}$$

with $m \in \mathbb{Z}\,\&\,\lambda = 1/2 + i\nu$. These states furnish an irreducible representation by construction. The conformal operators $\Theta_m^{(\lambda)}$ and $(\Theta_m^{(\lambda)})^\dagger$ are interpreted as creation and annihilation operators repectively and are used in expressing quantised fields expansions.

Building late-time representations with a field operator which inherits the conformal operator's quantum number from the late-time action of the generators (29) [26], the Casimir operator (30) can be expressed on functions of (τ, θ) as

$$\Box_g \Phi(\tau, \theta) = \lambda(1-\lambda)\Phi(\tau, \theta)\,. \tag{34}$$

By analogy to the Klein-Gordon equation (which we will talk about in greater detail in sections 3 and 4), $\Box_g \Phi(\tau, \theta) = m^2\ell^2\Phi(\tau, \theta)$, we can define $\lambda(1-\lambda) \equiv m^2\ell^2$, where $\lambda = 1/2 + i\nu$. As such, the dimension of the representation is linked to the mass of the theory and the massless case can be studied by analytical continuation to the Complementary Series and taking $\lambda \to 0$. This will be relevant when discussing de Sitter invariant massless vacuum states.

We now move on to the notion of observables in de Sitter spacetime before we look at the QFT.

2 Observables in de Sitter

Before naively starting to discuss fields in de Sitter spacetime and compute quantities, it is essential to define a notion of an observable in this particular framework and their significance. Indeed, working in a curved background usually means letting go of most of the usual notions we are able to define in Minkowski space. As we saw earlier, the absence of energy conservation ($\partial\tau$ Killing vector) in de Sitter is one of the main issues in defining observables. Indeed, the modes with positive frequency which are used to define vacuum states and Fock states are eigenfunctions of this Killing vector in Minkowski space [9]. In the absence of a $\partial\tau$ Killing vector, we now have to review the way in which we make a choice (or choices) for a vacuum state, since we don't have a canonical way of defining it.
In this section we review some of the necessary preliminary considerations when moving from a flat to a curved background for the study of fields, before looking at general quantum field theory in curved backgrounds.

2.1 Classical Considerations When Defining Observables in de Sitter Spacetimes

We start by looking at considerations that arise already at the classical level, when defining a boundary value problem for de Sitter space. Defining a boundary value problem is essential since we aim to describe the behaviour of fields in curved spacetime, which includes their behaviour at the boundary (should it be defined). We briefly review the Cauchy Problem and the Nulll Time-like Initial Value Problem, then we look at the problem with the concept of a physical boundary in de Sitter space, which is commonly used for placing boundary conditions on fields.

2.1.1 Cauchy Problem

Formally, the Cauchy Problem refers to the process of imposing data subject to constraints on hypersurface Σ_0 embedded in the manifold ($\equiv$ spacetime) and evolving it in time using Einstein's equations $G_{\mu\nu}$[1,10]. It has been proven that there exists a maximal development for any system of equations for which solutions define a hyperbolic manifold and for which the Cauchy Problem has a unique solution, up to isometry, in a neighbourhood of the original Σ_0 [10]. In de Sitter, the global geometry is a maximal development of an initial Cauchy surface. It is important to note that given two Cauchy surfaces Σ_0 and Σ_0' in the same

spacetime, their maximal development are isometric [10]. Therefore the choice of a Cauchy surface is irrelevant and can be chosen appropriately with regards to the problem at hand, since the results are isomorphic.
Less formally, we foliate our de Sitter spacetime with spacelike Cauchy surfaces (on which we will define the vector space to define fields) and evolve it in time. The foliation Σ_0 has metric $h_{\mu\nu}$, extrinsic curvature $k_{\mu\nu}$ and a time-like unit normal vector n^μ, such that

$$h_{\mu\nu} = g_{\mu\nu} + n_\mu n_\nu, \tag{35}$$

where $g_{\mu\nu}$ is the metric defined by the line element of de Sitter space (see eq. 2 or eq. 4 for example) [1].

2.1.2 Null Time-like Initial Value Problem

If we recall the geometric considerations of Section 1.2, we observed that a single observer cannot access the whole geometry of de Sitter space. This means that a consequential part of the data on Σ_0 will never be in causal contact with the single observer (if we assume that there is no exit to inflation and that the Universe is de Sitter everywhere [1]), which drives us to question the physical relevance of this data. Any observables that we could define outside of the worldtube of the observer is virtually irrelevant to any observation that could be made. We can partition Σ_0 between its observable part and unobservable part, respectively Σ_{obs} and Σ_{un}. We will not be treating our fields on this partition of the Cauchy slice as we consider them to be permeating the whole geometry of de Sitter space, but it is important to keep this question of relevance in mind.

2.1.3 Topological boundary consideration

One last consideration is the absence of a topological spatial boundary on de Sitter space. Indeed, if we recall the basic geometry of global de Sitter, it is the maximally symmetric globally hyperbolic solution to the Einstein equation with positive Cosmological Constant, with each constant τ slice being a 3-Sphere. This means that there is no spatial boundary on de Sitter space, which is a rather disturbing notion for imposing boundary conditions (such as Dirichlet or Neumann). The only boundary available to us is $\mathcal{I}^{\pm}$, and imposing boundary conditions on the infinite past or future proves rather troublesome [2]. Indeed if we try to impose a Dirichlet boundary at $\mathcal{I}^{\pm}$, fluxes are ruled out at the boundary, which means that conserved charges (like mass or momentum) do not carry over the boundary, but from a physical point of view, these charges don't vanish (although the absence of energy conservation on de Sitter seems to fit this picture), we are left with a conceptual paradox. We are also not usually inclined to fix the future.

One way this issue has been tackled, in the case of fields decaying at large infinity ie. $\phi \to 0$ as $\tau \to -\infty$, is by analytically continuing de Sitter spacetime to a 4D sphere through the coordinate transform $\tau \longrightarrow -i\tau$. The infinitely expanding de Sitter space becomes a compact sphere in 4-dimensional Euclidean spacetime and we can impose a smoothness condition across the southern or northern hemisphere to match conditions across $\mathcal{I}^-$ or $\mathcal{I}^+$ respectively [1,12,26]. We will come back to this in more detail when defining different vacuum states on de Sitter spacetime and when we look at the concept of a "Wavefunction of the Universe" [12].

2.2 Measurement Theory in de Sitter : a Fundamental Issue with Observables ?

Measurement Theory in de Sitter Space is a "recurrent nightmare" [13] for theoretical physicists. In General Relativity, we have to abandon the concept of local observables, as "attempting to collect too much information will eventually cause a gravitational backreaction "[1]. In other words, attempting to define too many observables in a local region of space has dramatic results. This why the concept of asymptotical symmetries and observables at asymptotic regions of space are fundamental to measurement in curved spacetimes. That way, we can define an S-matrix living at null infinity of these asymptotic regions, which is quite practical for studying fields in curved spacetimes [1]. However, as we saw in Section 2.1.3, de Sitter posesses no such asymptotic region, from a global topological perspective, as the spacelike region does not possess a boundary, and considering the fact that observers are confined to a thermal cavity, surrounded by a "finite sized cosmological horizon" [1], there is an impossibility to define observables in these regions like we do in flat space of Anti-de-Sitter (AdS) space.

In addition to that, we also have trouble defining precise observables on de Sitter. Indeed, de Sitter can be described by a finite Hilbert Space - this idea is debated, due to the non-compactness of the de Sitter group as we will see in §7 where we discuss Holography, and note we have only seen infinite-dimensional representations so far. This brings an irreducible inaccuracy to any quantum measurements and implies a non uniqueness of the quantum theory [13,43], as we will see when defining vacuum states in this background. We now proceed to reviewing the general theory of quantum fields in curved spaces.

3 General Quantum Field Theory in Curved Space

The basis of Quantum Field Theory in any background is defining a vacuum state from which a Fock space can be constructed using the mode expansion of fields, specifically positive frequency modes.

The usual procedure to do so is by quantising a scalar field by promoting it to an operator, expanding it in term of its modes and creation/annihilation operators. However, it is important to note that this is physically backwards. Indeed the quantum theory is the fundamental one and the "real world" is a classical limit of the quantum theory that we observe at our macroscopic scale .There are alternative procedures, like defining an algebra for the quantum observables [15], however, the procedure that we follow is a more direct approach which is a better fit for the purposes of this work. We will look at the algebra upon defining a Hilbert space from unitary irreducible representations.

The procedure to quantise fields is fairly similar in Minkowski QFT and curved space QFT, although the major difference is in the construction and significance of the vacuum state and Fock space. Indeed, in a general curved space there is an inherent ambiguity in the definition of the vacuum state(s)[14]. In Minkowski space, there is a natural set of modes associated with the Cartesian coordinates system, which are eigenfunctions of the ∂_τ Killing vectors, from which we can define the vacuum using the positive frequency modes [9] (we refer the reader to [9] for an overview of QFT in Minkowski space to contrast with curved space QFT). This natural choice of vacuum is of course not unique, however, it benefits from being invariant under the Poincaré group, and so this canonical vacuum is the "agreed vacuum" for all inertial frames [9].

General curved spaces will usually not benefit from such a symmetry group under which the vacuum is invariant. As we will see further on, de Sitter space has multiple coordinate systems under which we can define a "natural" set of modes from which we can define the vacuum state and Fock state, some of which do benefit from invariance under the de Sitter group. However, we will see that this invariance does not mean that all inertial frames will register the vacuum as a "no-particle state", which puts in question the relevance of this "vacuum".

It is worth noting as well that quantisation should not be attempted unless the set of coordinates on the manifolds covers the whole space , or does so upon maximal analytic extension or geodesic completeness . Although QFT covers phenomena on a microscopic local scale, the decomposition of fields into modes involves necessarily "global integral transformations" [14]. This is why global coordinates or planar coordinates are fit to study the quantisation of fields.

We will now review the general theory of quantum fields in curved spacetime to give a formal mathematical understanding before specifying to de Sitter spacetime.

Here, we broadly follow the procedure given by Birrel & Davies [9], which gives a very concise overview of the approach to QFT in curved spacetimes.

3.1 Classical Field Solutions

We start with the Lagrangian density

$$\mathcal{L}(x) = -\frac{1}{2}[-g(x)]^{1/2}\{g^{\mu\nu}(x)\partial_\mu\phi(x)\partial_\nu\phi(x) + [m^2 + \xi R(x)]\phi^2\}, \tag{36}$$

where $g(x)$ is the determinant of the metric $g_{\mu\nu}$, ξ is the gravitational coupling constant and $R(x)$ is the Ricci curvature. We confine our discussion to fixed spacetime, so $g(x)$ and $R(x)$ are taken as fixed quantities (independent of position). To that end, the matter coupling term can be absorbed into an effective mass term, which we will label m as well, without loss of generality. The theory will then be equivalent to the minimally coupled case, where $\xi = 0$. Another interesting case is the conformally coupled case, where the coupling constant is given by

$$\xi = \frac{1}{4}\left[(n-2)/(n-1)\right] = \frac{1}{6}, \tag{37}$$

where we have taken $n = 4$. The particularity of these cases is that the field equations become invariant under conformal transformation [9,16].

The action is given by

$$S = \int \mathcal{L}(x) d^4x, \tag{38}$$

and by requiring a null variation of the action with respect to ϕ, ie. $\delta S = 0$, we obtain the Klein Gordon equation in curved space as

$$\left[\Box_g - m^2\right]\phi(x) = 0, \tag{39}$$

where $\Box_g \equiv g^{\mu\nu}\nabla_\mu\nabla_\nu = \frac{1}{\sqrt{-g}}\partial_\mu\left[\sqrt{-g}g^{\mu\nu}\partial_\nu\right]$ is the covariant d'Alembertian operator in curved spacetime.

We define the scalar product on a Cauchy slice of the global spacetime (see Section 2.1.1) as

$$\langle \phi_1, \phi_2 \rangle = -i \int_{\Sigma} \phi_1 (x) \overleftrightarrow{\partial_\mu} \phi_2^* (x) \sqrt{-h} \mathrm{d}\Sigma^\mu , \tag{40}$$

where $d\Sigma^\mu = d\Sigma n^\mu$ is the volume element on the Cauchy slice Σ, and h is the determinant of the induced metric $h_{\mu\nu}$ defined in equation (35). As we said before, the choice of Σ is irrelevant since the results on Cauchy slices of a same space are isomorphic. As a result, the value of $\langle \phi_1, \phi_2 \rangle$ is independent of the choice of Σ [8].

There exists a complete orthonormal set of mode solutions u_i to the Klein Gordon equation. They satisfy the following conditions

$$\langle u_i, u_j \rangle = \delta_{ij} \ \ , \ \ \langle {u_i}^*, {u_j}^* \rangle = -\delta_{ij} \ \ , \ \ \langle u_i, {u_j}^* \rangle = 0 \tag{41}$$

where the index i is used to represent the necessary quantities to label the modes.

3.2 Quantisation

We then quantise the field expansion as a function of the modes by promoting the field to an operator and define the conjugate momentum in the usual way

$$\begin{aligned} \phi (x) &= \sum_i \left[a_i u_i (x) + a_i^\dagger u_i^* (x) \right] , \\ \Pi(x) &= \frac{\partial \mathcal{L}}{\partial(\partial_0)\phi} = -[-g]^{\frac{1}{2}} g^{00} \partial_0 \phi \ , \end{aligned} \tag{42}$$

where a_i and $a_i^\dagger$ are respectively the annihilation and creation operators with respect to the curved background. They obey the usual commutation relations

$$[a_i, {a_j}^\dagger] = \delta_{ij} \ , \tag{43}$$

and all other commutators vanish. The vacuum is constructed from these operators in the same way as for Minkowski, ie. $a_i|0\rangle = 0 \ , \forall i$, and the Fock space by acting on the vacuum state with the creation operator.

However, this is where the ambiguity in the formalism arises. The modes and vacuum state not being canonically defined, due to different non conformally isomorphic coordinate systems existing on the spacetime, we can obtain a different complete orthonormal set of modes solutions to the Klein Gordon equation, from which we can expand the field as

$$\phi(x) = \sum_i \Big[\bar{a}_i \bar{u}_i(x) + \bar{a}_i^\dagger \bar{u}_i^*(x)\Big]. \tag{44}$$

This allows us to define a new Fock state $\bar{a}_j|\bar{0}\rangle = 0$, $\forall j$ and consequently a new Fock space. As both sets of modes are complete and orthonormal, we can express them in terms of each other by means of a Bogoliubov transformation as

$$\bar{u}_j = \sum_i (\alpha_{ji} u_i + \beta_{ji} u_i{}^*) \ , \ u_j = \sum_i (\alpha_{ji} \bar{u}_i - \beta_{ji} \bar{u}_i^*) \ , \tag{45}$$

where the Bogoliubov coefficients α_{ji} and β_{ji} are evaluated as

$$\alpha_{ij} = \langle \bar{u}_i, u_j \rangle \ , \beta_{ij} = -\langle \bar{u}_i, u_j{}^* \rangle \ . \tag{46}$$

The fundamental operators can be expressed in terms of those transformations as well by equating the two expressions for the same field operator (see equations (42) and (44)), which gives

$$a_i = \sum_j (\alpha_{ji} \bar{a}_j + \beta_{ji}{}^* \bar{a}_j^\dagger) \ , \ \bar{a}_j = \sum_i (\alpha_{ji}{}^* a_i - \beta_{ji}{}^* \bar{a}_i^\dagger) \ . \tag{47}$$

The Bogoliubov coefficients satisfy the following properties

$$\begin{aligned} \sum_k (\alpha_{ik} \alpha_{jk}{}^* - \beta_{ik} \beta_{jk}{}^*) &= \delta_{ij} \ , \\ \sum_k (\alpha_{ik} \beta_{jk} - \beta_{ik} \alpha_{jk}) &= 0 \ . \end{aligned} \tag{48}$$

This allows us to show that the Fock spaces defined by the two different sets of modes solutions are different (for $\beta_{ij} \neq 0$), ie.

$$a_i |\bar{0}\rangle = \sum_j \beta_{ji}{}^* |\bar{1}_j\rangle \neq 0 \ . \tag{49}$$

Naturally, upon defining the number operator in our original set of modes as $N_i = a_i^\dagger a_i$ and computing its expectation value in our alternative vacuum sate, we get

$$\langle \bar{0} | N_i | \bar{0} \rangle = \sum_j |\beta_{ji}|^2 \ , \tag{50}$$

where we see a non-zero particle number expectation in our secondary vacuum state. Which is what allows us to question the relevance of this notion of vacuum state in a curved spacetime.

We now move on to look at fields in de Sitter spacetime.

4 Quantum fields in de Sitter Space

In this section, we specify the study of QFT to de Sitter spacetime where we will look at the quantisation of a free scalar field and the definition of a vacuum state in this background. For the vacuum state, we will look at different ways it can been defined and their significance. From this we will then be able to compute the two-point Correlation Function (Green Function) for a free scalar field.

Pure de Sitter space is a maximally symmetric globally hyperbolic spacetime with 10 degrees of symmetry, which is the same as Minkowski spacetime (Poincaré group and de Sitter group have the same dimensionality) [9]. Even though this greatly simplifies the study of QFT compared to general curved spacetimes, we still have to account for the additional considerations for the quantisation of fields introduced in the previous section which come with being in a curved spacetime.

We now proceed to the quantisation of a free scalar field in de Sitter and look at defining a vacuum state in planar and global coordinates (equations (4) and (2)).

4.1 Fields and Vacuum State in Planar Coordinates

If we recall, the metric in planar coordinates was given as $ds^2 = -d\tau^2 + e^{\frac{2\tau}{\ell}} d\vec{x}^2$. We can introduce a different conformal time coordinate as $\eta = -\ell e^{-\tau/\ell}$, which yields the metric with an overall ℓ^2 factor, which will simplify some algebraic manipulations.

$$ds^2 = \frac{\ell^2}{\eta^2}(-d\eta^2 + d\vec{x}^2)\,. \tag{51}$$

The Klein Gordon equation (see eq. 39) in this background can be expressed as (see Appendix B for details)

$$(\eta^2 \partial_\eta \eta^{-2} \partial_\eta + \overrightarrow{\partial_x}^2)\Phi = m^2 \ell^2 \Phi\,. \tag{52}$$

We can separate the variables of this equation in a general solution by expanding the field in a Fourier basis as $\Phi = \int \frac{d^3\vec{k}^2}{(2\pi)^3} e^{i\vec{k}.\vec{x}} \eta \tilde{\phi}(\eta)$. This yields the wave equation

$$(\eta^2 \partial_\eta \eta^{-2} \partial_\eta + \vec{k}^2)\eta\tilde{\phi}(\eta) = \eta^{-1} m^2 \ell^2 \tilde{\phi}(\eta) \tag{53}$$

We can solve this equation as a linear combination of two independent solutions expressed in terms of Bessel functions, namely

$$\begin{aligned}\tilde{\phi}_1(\eta) &= \eta^{1/2} J_\nu(k\eta),\\ \tilde{\phi}_2(\eta) &= \eta^{1/2} Y_\nu(k\eta)\,,\end{aligned} \tag{54}$$

with

$$\nu \equiv \sqrt{\frac{9}{4} - m^2\ell^2} \tag{55}$$

These solutions can be normalised with respect to the Klein-Gordon product, defined by equation (40) and conditions (41), ie.

$$\langle\tilde{\phi}_n, \tilde{\phi}_m\rangle = -i \int\limits_{\Sigma} \tilde{\phi}_n(x) \overleftrightarrow{\partial_\mu} \tilde{\phi}^*_m(x) \sqrt{-h} \mathrm{d}\Sigma^\mu = \delta_{nm}, \tag{56}$$

where the Cauchy slice Σ for this space can be taken as a constant time slice of the planar geometry, which is just a flat 3-slice, with the volume element $\mathrm{d}\Sigma = \mathrm{d}^3x$. The induced metric h is that of flat Euclidean 3-space and $\sqrt{-h} = i$. The normalised solutions are given as

$$\begin{aligned} \tilde{\phi}_1(\eta) &= (\pi\eta)^{1/2} J_\nu(k\eta) \\ \tilde{\phi}_2(\eta) &= (\pi\eta)^{1/2} Y_\nu(k\eta). \end{aligned} \tag{57}$$

If we recall from Quantum Mechanics, the vacuum is defined with respect to the Euclidean vacuum, with modes which are equivalent to positive frequency solutions in Minkowski space, ie. a positive frequency oscillatory behaviour in the asymptotic limit, equivalent to solutions in Minkowski space [1,8,9,22]. The behaviour of Bessel functions is well known and that of a particular set of linear combinations, namely the Hankel functions, which have oscillatory asymptotic behaviour [18,22]. The particular field solution which is equivalent to positive frequency modes in Minkowski space can be expressed in terms of a Hankel function of the second kind, namely [1,9]

$$\varphi_{E,k}(\eta) \equiv (\tilde{\phi}_1 - i\tilde{\phi}_2) = (\pi\eta)^{1/2} H_\nu{}^{(2)}(k\eta). \tag{58}$$

Near $\mathcal{I}^-$, ie. in the $k|\eta| \to \infty$ region, this particular solution behaves [1], as

$$\varphi_{E,k}(\eta) \underset{\lim\limits_{k|\eta|\to\infty}}{\longrightarrow} \frac{1}{\sqrt{2k}} e^{-ik\eta}. \tag{59}$$

Figure 2 shows a graphic representation of $\varphi_{E,k}(\eta)$ in the massless case, $m^2 = 0$, $\nu = 3/2$, where we can see the oscillatory behaviour at infinity. We will later confine our discussion to the massless case, as it will yield some interesting results on the significance of the two-point function of a massless scalar field. Each mass will have its own vacuum state and associated Fock space, as well as its own observables.

We can thus quantise our field and express it as

$$\hat{\Phi}_E(\eta, \vec{x}) = \int\limits_{\mathbb{R}^3} \frac{d^3\vec{k}}{(2\pi)^3} \left[\varphi_{E,k}(\eta) e^{i\vec{k}.\vec{x}} a^E_{\vec{k}} + \varphi_{E,k}{}^*(\eta) e^{-i\vec{k}.\vec{x}} \left(a^E_{\vec{k}}\right)^\dagger\right]. \tag{60}$$

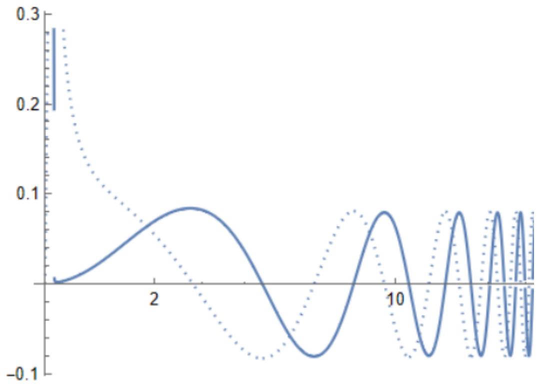

Figure 2: Plot of a non normalised massless φ_E solution. The hard blue line is the real part of the function and the dotted line the imaginary. We can see that as η increases, the solution becomes more and more oscillatory .

The creation and annihilation operators are specific to this field solution, and define their own vacuum state, and obey the canonical commutation relations

$$\begin{aligned} a^E_{\vec{k}} \left|\Omega\right\rangle &= 0, \ \forall \, \vec{k} \, , \\ \left[a^E_{\vec{k}} , \left(a^E_{\vec{k'}} \right)^\dagger \right] &= (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k'}) \, . \end{aligned} \tag{61}$$

We will come back to the significance and definition of this vacuum state when building a Hilbert space from unitary irreducible representations. We first look at the fields and vacuum in global coordinates.

4.2 Fields and Vacuum State in Global Coordinates

We will only be working with the planar geometry in the following sections, but we include the quantisation of fields in global coordinates to outline the difference in the constructed vacuum states. Recalling the metric given by equation (2), and following the same procedure as in the previous section, we obtain a Klein-Gordon equation which is not readily solvable by separation of variables, which is inconvenient to quantise the field and define a vacuum state. To overcome this, we can introduce a conformal time coordinate, to put it in a more convenient form. We define this conformal time as $T = \int\limits_{-\infty}^{\tau} \frac{d\tau}{\cosh(\tau/\ell)} = 2\tan^{-1}(e^{\tau/\ell})$, with $T \in [0, \pi)$. The metric becomes

$$ds^2 = \ell^2 \sin^{-2}(T) \left[-dT^2 + d\Omega_3{}^2 \right] , \tag{62}$$

where the 3-sphere metric is given by ${d\Omega_3}^2 = d\psi^2 + \sin^2(\psi)d\theta^2 + \sin^2(\psi)\sin^2(\theta)d\varphi^2$. From this metric, we can obtain the Klein-Gordon equation as (see Appendix B for details)

$$\left[-\partial_T \sin^{-2}(T)\partial_T + \sin^{-2}(T)\nabla^2_{S_3}\right]\Phi = \sin^{-4}(T)m^2\ell^2\Phi\,, \tag{63}$$

with $\nabla^2_{S_3}$ the Laplacian on the 3-sphere. This Laplacian has well known eigenmodes and eigenvalues [19] and we can express a general solution in terms of these modes known as Spherical Harmonics Y_{klm} with $\nabla^2_{S_3}Y_{klm} = -k(k+2)Y_{klm}$,

$$\Phi(T,\Omega) = \sum_{k,l,m} \Phi_k(T)Y_{klm}(\Omega). \tag{64}$$

By using this expression in equation (63), we obtain the wave equation as

$$\left[-\partial_T \sin^{-2}(T)\partial_T - \sin^{-2}(T)k(k+2)\right]\Phi_k(T) = \sin^{-4}(T)m^2\ell^2\Phi_k(T)\,, \tag{65}$$

for which we can solve directly for $\Phi(T)$. We obtain a solution as a linear combination of the two independent solutions expressed in terms of the Associated Legendre functions, namely

$$\begin{aligned}\Phi_{1,k}(T) &= C\big(-\sin^2(T)\big)^{3/4}P^{\nu}_{k+\frac{1}{2}}\left[\cos(T)\right]\\ \Phi_{2,k}(T) &= C\big(-\sin^2(T)\big)^{3/4}Q^{\nu}_{k+\frac{1}{2}}\left[\cos(T)\right]\,,\end{aligned} \tag{66}$$

with $P^{\nu}_{k+\frac{1}{2}}$ and $Q^{\nu}_{k+\frac{1}{2}}$ the Associated Legendre functions of the first and second kind respectively, C the normalisation constant and ν is the same as in equation (55). The solutions are normalised with respect to their respective Klein-Gordon product, ie. [27]

$$\int d\psi d\theta d\phi \sin^2(\psi)\sin(\theta)Y_{klm}(\Omega)Y^*_{klm}(\Omega) = \delta_{kk'}\delta_{ll'}\delta_{mm'} \quad , \tag{67}$$

which gives the normalisation constant as $C = \left[\frac{\pi}{4}\frac{\Gamma(k+3/2-\nu)}{\Gamma(k+3/2+\nu)}\right]^{1/2}e^{i(\pi/2)\nu}$ [27].

Contrary to the planar geometry case from §4.1, these solutions do not have a readily available oscillatory solution as $T \to \pi$ to associate a positive frequency mode to construct the vacuum. To look for a linear combination that satisfies the association to positive frequency modes, we analytically continue our space to a Euclidean four-sphere by taking the transformation $T \to -i\theta$. As mentioned in §2.1.3, our infinitely expanding de Sitter space becomes a compact four-sphere, and we can look for solutions which are smooth across the southern hemisphere to match the ground state requirements. This Euclidean vacuum will give us the vacuum state in de Sitter space upon Wick rotating back to our original coordinate system.

Upon transformation to Euclidean spacetime, our solutions (66) become

$$\begin{aligned} \tilde{\Phi}_{1,k}(T) &= \left[\frac{\pi}{4}\frac{\Gamma(k+3/2-\nu)}{\Gamma(k+3/2+\nu)}\right]^{1/2} e^{i(\pi/2)\nu}\sinh^{3/2}(T)P^{\nu}_{k+\frac{1}{2}}\left[\cosh(T)\right] \\ \tilde{\Phi}_{2,k}(T) &= \left[\frac{\pi}{4}\frac{\Gamma(k+3/2-\nu)}{\Gamma(k+3/2+\nu)}\right]^{1/2} e^{i(\pi/2)\nu}\sinh^{3/2}(T)Q^{\nu}_{k+\frac{1}{2}}\left[\cosh(T)\right] \ , \end{aligned} \tag{68}$$

and from these we can look for a solution which is smooth accross the southern hemisphere of our Euclidean four-sphere. This has already been solved, and we get our positive frequency solution as [9,20,27]

$$\Phi^E_k(T) = \tilde{\Phi}_{1,k}(T) + \frac{2i}{\pi}\tilde{\Phi}_{2,k}(T). \tag{69}$$

In a similar manner as in the planar geometry case, we can thus quantise and expand our field solution as

$$\hat{\Phi}(T,\Omega) = \sum_{k,l,m}\left[Y_{klm}(\Omega)\Phi^E_k(T)a^E_{k,l,m} + Y^*_{klm}(\Omega)\big(\Phi^E_k(T)\big)^*\big(a^E_{k,l,m}\big)^\dagger\right]. \tag{70}$$

Similarly, to the planar geometry case, we have our creation/annihilation operators from which we can construct our vacuum state,

$$\begin{aligned} &a^E_{k,l,m}\left|0\right\rangle = 0, \ \forall \ k,l,m\,, \\ &\left[a^E_{k,l,m}, \big(a^E_{k',l',m'}\big)^\dagger\right] = \delta_{kk'}\delta_{ll'}\delta_{mm'}\,. \end{aligned} \tag{71}$$

It is quite obvious from the construction of our two vacuum states that the creation and annihilation operators differ for each geometry. We can therefore infer, without having to perform the Bogoliubov transformations, that the vacuum defined in planar coordinates will not vanish under the annihilation operator associated with equation (69) and the vacuum defined in global coordinates will not vanish under the annihilation operator associated with equation (58).

Moreover, in a series of interesting results by Tagirov [9,20] and Gibbons & Hawking [9,21], they found that a comoving observer (conformal coupling eq. (37)) in the steady state of planar de Sitter space (ie. confined to $-\infty < \eta < 0$) and in global de Sitter space would perceive the massless vacuum as "non empty". Firstly, they found that the two-point Wightman function $G^+(x(\tau), x(\tau'))$ is the same for both vacuum states, which means that these vacuum states are equivalent, even though one would register them as non empty upon looking at them from the other's perspective. Secondly, they found that upon computing the Response Function $\mathcal{F}(E)$ (which, broadly speaking, describes the probability amplitude of a detector registering a particle) using the two-point Wightman function , they found a response to this vacuum state as

$$\begin{aligned}\mathcal{F}(E) &= \int_{-\infty}^{\infty} d\tau \int_{-\infty}^{\infty} d\tau' e^{-iE(\tau-\tau')} G^{+}(x(\tau), x(\tau')) \\ \frac{\mathcal{F}(E)}{\text{unit time}} &= (E/2\pi)(e^{2\pi\ell E} - 1)^{-1},\end{aligned} \tag{72}$$

where E is the energy of an arbitrary particle in the vacuum and ℓ the de Sitter length.
It turns out that this result corresponds to a thermal radiation spectrum with temperature $T = 1/(2\pi\ell k_B)$, where k_B is the Boltzmann constant, emanating from beyond the cosmological event horizon [21]. This suggests that the Universe's vacuum has a finite temperature and these results and the way they are derived suggest abandoning "the idea that particles should be defined in an observer-independent manner" [21]. Not only do we end up having trouble with the notion of a state empty of particles but we even have trouble with the concept of particles themselves (not that the latter is particularly new, but this is some additional complication) and we have a finite temperature for emptiness.
We will not go further into the details of Cosmological Particle Creation as it is beyond the scope of our discussion, but this aspect of the vacuum state is one of the most interesting results that emanated from studying quantum fields in de Sitter spacetime and it seemed necessary to mention it as it raises some fundamental conceptual questions.
One question we now ask ourselves, since we now know that this vacuum state doesn't seem to be an actual vacuum, is what exactly the precise nature of this vacuum state is. This is what we will try to answer now by looking at the Bunch-Davies state and the construction of the Hilbert space.

4.3 Bunch-Davies State and the Hilbert Space

The Bunch-Davies state is the vacuum state that we have just constructed in the previous section. It is a de Sitter invariant vacuum state [8,9,15,16,25]. It can be constructed in multiple ways, from the field solutions [16,20,23,25,27] like we have done, from a Bogoliubov transformation of the Euclidean vacuum [8], or by solving for two-point functions satisfying de Sitter invariance [15] for example. There is some trouble however in defining a Bunch-Davies state for the massless minimally coupled case as there does not exist a de Sitter invariant vacuum state [8,15,27]. Indeed when defining a vacuum state for arbitrary mass

m, we observe a breakdown of the invariance as $m^2 \to 0$. In reality there always subsits a small gravitational coupling acting as an effective mass and there will be an invariant vacuum state.

The Hilbert space is arranged in terms of unitary irreps. The Principal Series for dS_4 has $\lambda = 3/2 + i\nu$ and the Complementary Series has $\lambda \in (0, 3)$ in the spin 0 case [23,26]. Then given an invariant Bunch-Davies vacuum state $|\Omega\rangle$, we can build a Hilbert space by acting with the creation operators defined in (61) to build single particle states $\left(a^E_{\vec{k}}\right)^\dagger |\Omega\rangle \equiv |\lambda, k\rangle$ and multiparticle states by acting repeatedly. These states have the inherited normalisation from (61) [23]

$$\langle \lambda, k|\lambda, k'\rangle = (2\pi)^3 \delta^{(3)}(\vec{k} - \vec{k'}) \ . \tag{73}$$

The state $|\Omega\rangle$ is annihilated by all generators of the Lie algebra (given in equations (21) and (23)) and the rest of the $|\lambda, k\rangle$ states transform under the same infinitesimal generators [23]. In addition we can act on the states with the Casimir operator which in dS_4 we can express from the conformal algebra as [23]

$$C = D^2 - \frac{1}{2}(K_\mu P^\mu + P_\mu K^\mu + M_{\mu\nu} M^{\mu\nu}). \tag{74}$$

It acts on the states as

$$C|\lambda, k\rangle = \lambda(1 - \lambda)|\lambda, k\rangle \tag{75}$$

which is a spin-zero unitary irrep of SO(1,4)[23]. We recall as well that we have defined $\lambda(1 - \lambda) \equiv m^2\ell^2$. As such we can construct a massless vacuum state in the Complementary Series by taking the limit $\lambda \to 0$ which will be de Sitter invariant. We now have everything to move on to computing the two-point function of a massless vacuum state, which is an expectation value in the vacuum state we have defined here.

5 Two-point Function of the Massless Scalar Field

The two-point function is one of the most fundamental and simple observables in QFT. In this section, we compute the positive frequency Whightman function G^+ for a massless scalar field. This function is the positive frequency part of the Schwinger function iG, computed as the vacuum expectation value of the commutator of fields [9]

$$iG(x, x') = \langle 0|[\phi(x)\phi(x')]|0\rangle = G^+(x, x') - G^-(x, x'). \tag{76}$$

It is a correlation function of a scalar field and is computed by taking the vacuum expectation value of the first term in the commutator of fields at two different spacetime positions x and x'[9], ie.

$$G^{+}(x,x') = \langle 0|\phi(x)\phi(x')|0\rangle \,. \tag{77}$$

We are particularly interested in computing the late-time two-point function of a massless scalar field, as we will show how instructive the late-time behaviour of light fields is.
We start by deriving the general form from which the two-point function will be computed in planar coordinates. We first recall our quantised field solution $\hat{\varphi}_E(\eta,\vec{x})$ given by equation (60), we input that solution directly in (77) with the appropriate Bunch-Davies vacuum $|\Omega\rangle$, which gives

$$\begin{aligned}\langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle &= \\ &= \int\limits_{\mathbb{R}^3}\frac{d^3\vec{k}}{(2\pi)^3}\int\limits_{\mathbb{R}^3}\frac{d^3\vec{k'}}{(2\pi)^3}\langle\Omega|\left(\varphi_{E,k}(\eta)e^{i\vec{k}.\vec{x}}a^E_{\vec{k}} + {\varphi_{E,k}}^*(\eta)e^{-i\vec{k}.\vec{x}}\left(a^E_{\vec{k}}\right)^\dagger\right)\times \\ &\times\left(\varphi_{E,k'}(\eta)e^{i\vec{k'}.\vec{y}}a^E_{\vec{k'}} + {\varphi_{E,k'}}^*(\eta)e^{-i\vec{k'}.\vec{y}}\left(a^E_{\vec{k'}}\right)^\dagger\right)|\Omega\rangle\,.\end{aligned} \tag{78}$$

Considering we know the action of the annihilation operators on the vacuum (see eq. 61)), and all states are orthogonal (hence terms with different number of annihilation and creation operators vanish), we can readily see that the only non vanishing part of the above equation is the $a^E_{\vec{k}}\left(a^E_{\vec{k'}}\right)^\dagger$ crossterm. Our two-point function becomes

$$\begin{aligned}&\langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle = \\ &= \int\limits_{\mathbb{R}^3}\frac{d^3\vec{k}}{(2\pi)^3}\int\limits_{\mathbb{R}^3}\frac{d^3\vec{k'}}{(2\pi)^3}\langle\Omega|\varphi_{E,k}(\eta){\varphi_{E,k'}}^*(\eta)e^{i\vec{k}.\vec{x}}e^{-i\vec{k'}.\vec{y}}a^E_{\vec{k}}\left(a^E_{\vec{k'}}\right)^\dagger|\Omega\rangle \\ &= \int\limits_{\mathbb{R}^3}\frac{d^3\vec{k}}{(2\pi)^3}\int\limits_{\mathbb{R}^3}\frac{d^3\vec{k'}}{(2\pi)^3}\langle\Omega|\varphi_{E,k}(\eta){\varphi_{E,k'}}^*(\eta)e^{i\vec{k}.\vec{x}}e^{-i\vec{k'}.\vec{y}}\left[a^E_{\vec{k}},\left(a^E_{\vec{k'}}\right)^\dagger\right]|\Omega\rangle\,.\end{aligned} \tag{79}$$

Note that in the second line, we are able to replace the $a^E_{\vec{k}}\left(a^E_{\vec{k'}}\right)^\dagger$ term with the commutator, as the second term of the commutator will vanish due to the annihilation operator acting on the vacuum. Using the commutator of the operators (see eq. (61)), our above equation becomes

$$\langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle = \int\limits_{\mathbb{R}^3}\frac{d^3\vec{k}}{(2\pi)^3}\varphi_E\varphi_E^* e^{i\vec{k}.(\vec{x}-\vec{y})}\,. \tag{80}$$

This last expression is the continuous analog of the mode sum computed by
Bunch & Davies for the two-point function [9,16]. The reason for which we have a continuous

integral is that we are working with a continuous set of modes $\vec{k}$. This can be seen in the separation of variables over the modes, we have a continuous Fourier expansion in planar coordinates. Bunch&Davies are working with a discrete set of modes [16] (as in [1]). In global coordinates the separation of variables in terms of Spherical Harmonics leads to a dicretisation of modes as well. See for example [27] for the Hadmard's elementary function (correlation function of the anticommuator of fields) calculated as a mode sum in global coordinates.

We can now compute the value of this two-point function using our solutions from §4.1. In the massless case, equation 58 reduces to (after normalisation)

$$\varphi_{E,k}(\eta) = \frac{1}{k^{3/2}}(1 - ik\eta)e^{ik\eta}. \tag{81}$$

To study the late-time behaviour of our two-point function, we expand the field near $\mathcal{I}^+$, which corresponds to the $\eta \to 0$ region. We get the field near $\eta = 0$ as

$$\varphi_{E,k}(\eta) \underset{\lim_{\eta \to 0}}{\longrightarrow} \frac{1}{k^{3/2}} \tag{82}$$

and so we get the late-time two-point function as

$$\lim_{\eta \to 0} \langle \Omega | \hat{\Phi}_E(\eta, \vec{x}) \hat{\Phi}_E(\eta, \vec{y}) | \Omega \rangle = \int \frac{d^3\vec{k}}{(2\pi)^3} \frac{1}{k^3} e^{i\vec{k}.(\vec{x}-\vec{y})} . \tag{83}$$

By expressing $\vec{k}$ in the usual momentum space sphere basis (k, θ, ϕ) , we can express the volume element as $d^3\vec{k} = dkd(\cos(\theta))d\phi$ and we can express the exponential term as $e^{ikr\cos(\theta)}$ where $r \equiv |\vec{x} - \vec{y}|$. From that we can express (83) as

$$\begin{aligned} \lim_{\eta \to 0} \langle \Omega | \hat{\Phi}_E(\eta, \vec{x}) \hat{\Phi}_E(\eta, \vec{y}) | \Omega \rangle &= \int_0^\infty \frac{dk}{(2\pi)^3} \int_0^\pi d(\cos(\theta)) \int_0^{2\pi} d\phi \frac{1}{k^3} e^{ikr\cos(\theta)} \\ &= -\frac{2}{(2\pi)^2} \int_0^\infty \frac{dk}{k} \frac{\sin(kr)}{kr} . \end{aligned} \tag{84}$$

This expression has an IR divergence (the integral diverges as $k \to 0$), which stems from the fact that we are using a massless minimally coupled vacuum state, which is not de Sitter invariant [23,27]. There are a few ways we can deal with this it to obtain a physical result. The more mathematical approach is to do it through analytic continuation [29,30], which is a rather rigorous process to regularise the divergence. For our purpose we opt for a more physically motivated method. As k approaches zero, the modes become unphyiscal, as the wavelengths become increasingly large, larger than the de sitter length ℓ. To set a physical limit for our modes, we can set a lower cutoff $\epsilon > 0$ to the values over which k is integrated such that the wavelength of the field $\lambda_f \ll \ell$ (not to be mistaken with λ the dimension of the irreducible representation). Our integral becomes

$$\lim_{\eta\to 0} \langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle = -\frac{1}{2\pi^2}\int_{\epsilon}^{\infty}\frac{dk}{k}\frac{\sin(kr)}{kr} \ , \tag{85}$$

which we compute as

$$\lim_{\eta\to 0} \langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle = \frac{1}{2\pi^2}\left[Ci(r\epsilon) + \frac{\sin(r\epsilon)}{r\epsilon}\right] \ , \tag{86}$$

where $Ci(r\epsilon)$ is the Cosine Integral defined as $Ci(x) = -\int_x^{\infty} \frac{\cos(t)dt}{t}$ defined over $\mathbb{R}$ ($Ci(z)$ can have a complex argument and is defined differently), as we take r and ϵ real-valued. We can expand $Ci(z)$ as a series to the first order which gives our final result for the late-time two-point function as

$$\lim_{\eta\to 0} \langle\Omega|\hat{\Phi}_E(\eta,\vec{x})\hat{\Phi}_E(\eta,\vec{y})|\Omega\rangle = \frac{1}{2\pi^2}\left[1+\gamma+\ln(\epsilon|\vec{x}-\vec{y}|)\right] + \mathcal{O}(r^2) \ , \tag{87}$$

with γ the Euler-Mascheroni constant.

This result, even though it is not new, is quite remarkable. Indeed the two-point function for a massless scalar field has been computed before through different methods, like restricting the massive solution, which is expressed in terms of a hypergeometric function, to the massless case, and there have been more rigorous computations, through analytical continuation for example. We refer the reader to [16,27,31,32] for some prominent examples. It should be noted that these results are not computed in the late-time limit. For a late time-limit two-point function, [23] gives a concise computation from the action of boundary conformal operators (as defined in §1.4), which can then be taken to the massless limit as $\lambda \to 0$ [26] and yield a similar result.

What is remarkable about the late-time limit, is that we see that the fields never decorrelate, which is in strong opposition with the Cluster Decomposition Principle [26,35]. Indeed, this principle implies that interactions should be local, and that interacting fields should decorrelate given a sufficiently large amount of time or large enough spatial distances [33], which is not observed here. It turns out as well that this logarithmic behaviour corresponds to the temperature inhomogeneities observed in the CMB, referred to as Gaussian and scale invariant spectrum [26,36,37]. This correlates the observations that the CMB is at thermal equilibrium, with similar inhomogeneities, across the whole spectrum in parts that seem like they were never in contact (coming from different region of space). This is one of the consequences of inflation.

This means that from this simple two-point massless function in the late-time limit, we obtain a strong but not definitive argument in favour of an inflationary de Sitter model of the Universe. There is still work to be done in order to conclude on accepting this theory, but this result certainly constitutes a strong motive to direct efforts into this direction.

For the rest of this discussion, we review some more conceptual aspects of de Sitter space, and we start with the Wavefunction of the Universe.

6 Wavefunction of the Universe : An Initial State ?

"A physics student is taking a history test and is asked to identify two causes of World War I. In the essay space on the test paper, the student writes, 'The universal wave function, and the initial and boundary conditions of the universe.'" [38]

This may have been written as a joke, but the wavefunction of the universe and its intitial boundary conditions is a concept that Hartle and Hawking took to heart [12]. They proposed a wavefunctional formalism of quantum gravity from a Schrödinger picture dynamics with an initial state as a ground state defined from the Euclidean vacuum.

If we recall from quantum mechanics, we have a Schrödinger equation $i\partial_t\psi = H\psi$ for which we can naturally express a ground state as a path integral Wick rotate to Euclidean time

$$\psi_0(x, t=0) = \mathcal{N}\int \delta x(t) e^{-I[x(t)]} \tag{88}$$

where $I[x(t)]$ is the Euclidean action obtained from the Lorentzian action S by Wick rotation. Their proposal for quantum gravity is a solution to the Wheeler-DeWitt equation [12] which is analogous to the Schrödinger equation but for a wavefunctional in a theory of gravity. The wavefunctional state Ψ is interpreted as the amplitude of a three-dimensional metric h_{ij} forming the boundary of a four-dimensional spacetime[1,12,50]. In a closed universe, with the wavefunctional written as a function of the three-metric of a scalar field ϕ $\Psi[h_{ij}, \phi]$, the Wheeler-DeWitt equation is derived [12] as

$$\left\{G_{ijkl}\frac{\delta^2}{\delta h_{ij}\delta h_{kl}} + h^{\frac{1}{2}}\left[-{}^3R(h) + 2\Lambda + l^2 T_{nm}\left(-i\frac{\delta}{\delta\phi}, \phi\right)\right]\right\}\Psi[h_{ij}, \phi] = 0\,, \tag{89}$$

where G_{ijkl} is the metric on the superspace defined by

$$G_{ijkl} = \frac{1}{2} h^{-1/2} (h_{jk} h_{jl} + h_{il} h_{jk} - h_{ij} h_{kl}) \,, \tag{90}$$

${}^3R$ is the "scalar curvature of the intrinsic geometry of the three-surface" and l is the Planck length $l = (16\pi G)^{1/2}$ with our units $\hbar = c = 1$ [12].

Without going into the details, they found the ground state of their solution to be de Sitter spacetime in the classical limit. From their proposal they obtain a de Sitter geometry as an initial state of the Universe through this wavefunctional formalism of quantum gravity.

The exact significance of this result, outside of inflationary implications for the Universe and representing motivation, is outside of our expertise and the wavefunctional formalism for describing quantum gravity in de Sitter is still an active area of research - see [51] for an example of this formalism applied to the late-time structure of the Bunch-Davies state of light fields and quantum gravity.

In the next section we review the search for a holographic description of de Sitter spacetime.

7 Holography in de Sitter: dS/CFT Correspondence

The idea for a holographic principle in de Sitter spacetime is analogous to the AdS/CFT case. This correspondence is deduced from the isometry group of five-dimensional AdS being dual to the boundary Super Yang-Mills theory with the conformal group O(2,4) [43,44]. In the dS case, there would be a dual boundary theory with conformal symmetry SO(1,4). However de Sitter, contrary to AdS spacetime, has cosmological horizons, with finite entropy [21,43]. This could imply a finiteness in the Hilbert description of the causal patch as an isolated quantum system, which is necessary to a dual formulation of the holographic theory [44].

However this raises some issues, which have led to claims that there cannot be a formulation of a holographic dual to de Sitter space. Some of these involve the non-compactness of the de Sitter group and the fact that only compact groups have unitary nontrivial finite-dimensional representations [11,43], or the fact that most generators from the de Sitter group transform the causal patch to another causal patch, which implies that they do not act on the Hilbert of a single observer [44]. This led to the concept of observer compeimentarity [43,44].

Weakly-coupled theories are well studied and there have recently been examples of strongly coupled CFT's in de Sitter via holographic theory (see [45] for example). There also has been a conjecture of dS/CFT proposed which is formulated as a one-to-one correspondence between "the boundary-to-boundary correlation functions at $\mathcal{I}^+$ with future boundary conditions of $(d+1)$-dimensional de Sitter spacetime and the correlation functions of a d-dimensional Euclidean CFT" [1,46,47].

If we consider the late-time behaviour of a free scalar field we have a direct analogy with the AdS case. The field of mass m in planar coordinates behaves as [1]

$$\Phi(\eta,\vec{x}) = \eta^{\Delta_m^+}\left(A(\vec{x}) + \mathcal{O}(\eta^2)\right) + \eta^{\Delta_m^-}\left(B(\vec{x}) + \mathcal{O}(\eta^2)\right) , \tag{91}$$

where the operator dual to Φ has conformal weight

$$\Delta_m^{\pm} = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} - m^2\ell^2}\,. \tag{92}$$

The choice of $\Delta_m^{\pm}$ is fixed by the choice of future boundary conditions. Notice that for large enough masses, the conformal weight is complex, which implies that the CFT may be non-unitary and the presence of complex operators [1,46]. Holography in de Sitter space remains an open problem with the search for a holographic description of quantum gravity or string theory in Asymptotically de Sitter spacetimes.

This brings us to our last subject - String Theory in de Sitter.

8 Strings in de Sitter

In this section we very briefly review an interesting aspect of strings in de Sitter spacetime. Building a theory of strings in de Sitter represent its own area of research and there has been many attempts, with multiple challenges [48], which is beyond the scope of our discussion. We only mention here one challenge which seems appropriate with our earlier discussion about vacuum states in de Sitter.

Any formulation so far has not been able to define a stable vacuum. They all seem to give rise (to the best of our knowledge) to metastable vacuum states [1,48,49]. These are states with a non-vanishing probability of a transition to a more stable vacuum state with a lower energy. As stable vacua could be defined in Minkowski and AdS as supersymmetric solutions to string theory [1], the impossibility to define a stable vacuum in de Sitter might lie within a more fundamental, structural issue.

9 Discussion

We consider this article as a gateway to advanced research in de Sitter space. Considering that Asymptotic de Sitter spacetimes are becoming more relevant, the necessity of understanding their fundamental structure is increasing. After having reviewed the structure of the spacetime's geometry, algebra and representations, and challenges of this background in the context of quantising fields, we outlined the issues arising with accepted notions like symmetries, defining observables, and the vacuum state. We went through a procedure for obtaining the Bunch-Davies and briefly mentioned the cosmological particle creation point of view through a couple of important results for the significance of the Bunch-Davies and notion of vacuum. This allowed us to compute one of the most fundamental results of quantum fields in de Sitter space through the two-point function of a massless scalar field. Our computation of the two-point function is not new and although the approach is not the most rigorous, we nonetheless find the late-time logarithmic behaviour of light fields, corresponding to the gaussian scale invariant spectrum of the CMB. Even though one of the most fundamental observables of QFT being physically observed in the CMB is a consequential step forward, there are other theories that predict a similar observation. There remains a tremendous amount of work to be done which represents a rich area of research.

We only touched the surface of three main areas of research in section 6, 7 and 8. These three main areas are vastly more complex and there are other areas, some dealing with more structural challenges and they are beyond our expertise. There has recently been developments in infrared divergences [29,30,52] for example, which could lead to resolving some more fundamental issues.

There is also a problem with the prediction of the Cosmological Constant in QFT [53]. The predicted value is off by an order of $\sim 10^{120}$ from cosmological observations, which is dramatic for a theory that has empirically held up so well. To the best of our knowledge, there has not been yet a conclusive explanation for the origin of this problem and it has not had a successful attempt at regularisation, and it puts in question the so far accepted ΛCDM model.

There is no shortage of areas of research and challenges in de Sitter spacetime, and these furnish a very exciting field of theoretical physics for the coming future.

10 Appendices

Appendix A : Derivation of 2D de Sitter Killing Vectors

As stated previously, the hyperboloid is embedded in the following way, with the differentials associated to each embedding coordinate

$$
\begin{aligned}
&-\left(X^0\right)^2+\left(X^1\right)^2+\left(X^2\right)^2=\ell^2 \\
&X^0=\ell\sinh(\frac{\tau}{\ell})\ ;\ dX^0|_\Sigma=\cosh(\frac{\tau}{\ell})d\tau \\
&X^1=\ell\cosh(\frac{\tau}{\ell})\sin(\theta)\ ;\ dX^1|_\Sigma=\sinh(\frac{\tau}{\ell})\sin(\theta)d\tau+\ell\cosh(\frac{\tau}{\ell})\cos(\theta)d\theta \\
&X^2=\ell\cosh(\frac{\tau}{\ell})\cos(\theta)\ ;\ dX^2|_\Sigma=\sinh(\frac{\tau}{\ell})\cos(\theta)d\tau-\ell\cosh(\frac{\tau}{\ell})\sin(\theta)d\theta \\
&ds^2=-d\tau^2+\ell^2\cosh^2(\frac{\tau}{\ell})d\theta\,.
\end{aligned}
$$

The Killing vectors on the ambient space, solutions to equation (13) in 3 dimensional Minkowski space, are found as

$$
\begin{aligned}
&\delta\xi^\mu_{12}\partial_\mu=X^1\partial_2-X^2\partial_1 \\
&\delta\xi^\mu_{0\alpha}\partial_\mu=X^0\partial_\alpha+X^\alpha\partial_0\quad,\alpha=1,2\ .
\end{aligned}
$$

Using equation (18) by restricting and evaluating it on Σ, we can find the Killing vectors in global de Sitter coordinates on its geometry.We start by evaluating the $\delta\xi^\mu_{12}$ Killing vector.

$$
\begin{aligned}
&g^{ind}_{\mu v}\delta\xi^\mu_{\alpha\beta}dx^\nu=\eta_{AB}\delta\xi^A_{\alpha\beta}dX^B|_\Sigma \\
&\Rightarrow-\delta\xi^1_{12}d\tau+\ell^2\cosh^2(\frac{\tau}{\ell})\delta\xi^2_{12}d\theta=-\delta\xi^0_{12}dX^0|_\Sigma+... \\
&\quad=-X^2dX^1|_\Sigma+X^1dX^2|_\Sigma \\
&\quad=-\ell\cosh(\frac{\tau}{\ell})\cos(\theta)(\sinh(\frac{\tau}{\ell})\sin(\theta)d\tau+\ell\cosh(\frac{\tau}{\ell})\cos(\theta)d\theta) \\
&\quad+\ell\cosh(\frac{\tau}{\ell})\sin(\theta)(\sinh(\frac{\tau}{\ell})\cos(\theta)d\tau-\ell\cosh(\frac{\tau}{\ell})\sin(\theta)d\theta) \\
&\quad=-\ell^2\cosh^2(\frac{\tau}{\ell})d\theta
\end{aligned}
$$

By identification, we obtain the components of $\delta\xi^\mu_{12}$ on the hyperboloid in de Sitter coordinates and express the Killing vector field as

$$\delta\xi^\mu_{12}\partial_\mu=-\partial_\theta\,.$$

Similarly, we repeat the procedure for the remaining Killing vectors.

$$
\begin{aligned}
&-\delta\xi^1_{01}d\tau + \ell^2\cosh^2(\frac{\tau}{\ell})\delta\xi^2_{01}d\theta = -X^1dX^0|_\Sigma + X^0dX^1|_\Sigma \\
&= -\ell\cosh^2(\frac{\tau}{\ell})\sin(\theta)d\tau \\
&\ + \ell\sinh(\frac{\tau}{\ell})(\sinh(\frac{\tau}{\ell})\sin(\theta)d\tau + \ell\cosh(\frac{\tau}{\ell})\cos(\theta)d\theta) \\
&= \ell\sin(\theta)(-\cosh^2(\frac{\tau}{\ell}) + \sinh^2(\frac{\tau}{\ell}))d\tau + \ell^2\cosh(\frac{\tau}{\ell})\cos(\theta)\sinh(\frac{\tau}{\ell})d\theta
\end{aligned}
$$

From that, we get

$$
\begin{aligned}
\delta\xi^\mu_{01}\partial_\mu &= -\ell\sin(\theta)(-\cosh^2(\frac{\tau}{\ell}) + \sinh^2(\frac{\tau}{\ell}))\partial_\tau + \tanh(\frac{\tau}{\ell})\cos(\theta)\partial_\theta \\
&= \ell\sin(\theta)\partial_\tau + \tanh(\frac{\tau}{\ell})\cos(\theta)\partial_\theta
\end{aligned}
$$

And the last with

$$
\begin{aligned}
&-\delta\xi^1_{02}d\tau + \delta\xi^2_{02}\ell^2\cosh^2(\frac{\tau}{\ell})d\theta = -X^2dX^0|_\Sigma + X^0dX^2|_\Sigma \\
&= -\ell\cosh^2(\frac{\tau}{\ell})\cos(\theta)d\tau + \ell\sinh(\frac{\tau}{\ell})(\sinh(\frac{\tau}{\ell})\cos(\theta)d\tau - \cosh(\frac{\tau}{\ell})\sin(\theta)d\theta) \\
&= d\tau\ell\cos(\theta)(\cosh^2(\frac{\tau}{\ell}) + \sinh^2(\frac{\tau}{\ell})) - \ell\cosh(\frac{\tau}{\ell})\sinh(\frac{\tau}{\ell})\sin(\theta)d\theta
\end{aligned}
$$

From which, we obtain our last Killing vector field as

$$
\begin{aligned}
\delta\xi^\mu_{02}\partial_\mu &= -\cos(\theta)(-\cosh^2(\frac{\tau}{\ell}) + \sinh^2(\frac{\tau}{\ell}))\partial_\tau - \tanh(\frac{\tau}{\ell})\sin(\theta)\partial_\theta \\
&= \ell\cos(\theta)\partial_\tau - \tanh(\frac{\tau}{\ell})\sin(\theta)\partial_\theta
\end{aligned}
$$

Appendix B : derivation of Klein-Gordon equations

Klein-Gordon in Planar Coordinates

Deriving the Klein-Gordon equation in planar coordinates is rather straightforward but we show how we achieved it here for completeness.

Starting from our metric given by equation (51), we derive the necessary quantities to compute the Klein-Gordon equation as

$$\sqrt{-g} = \frac{\ell^2}{\eta^2} \quad , \quad g^{\mu\nu} = \frac{\eta^2}{\ell^2} \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

These quantities can be plugged directly into equation (39), which yields

$$(\frac{\eta^4}{\ell^4}\partial_\eta \ell^2 \eta^{-2} \partial_\eta + \frac{\eta^2}{\ell^2}\overrightarrow{\partial_x}^2)\Phi = m^2\Phi.$$

It is easily seen that we can rearrange from there to obtain the same equation as (52), namely

$$(\eta^2\partial_\eta\eta^{-2}\partial_\eta + \overrightarrow{\partial_x}^2)\Phi = m^2\ell^2\Phi$$

Klein-Gordon in Global Coordinates

The derivation of the Klein Gordon equation in global coordinates is not as straight forward as for the planar geometry. It involves some rearrangement as well, the nature of which is slightly more complex.

The full form of the metric is

$$ds^2 = \ell^2 \sin^{-2}(T)\left[-dT^2 + d\psi^2 + \sin^2(\psi)d\theta^2 + \sin^2(\psi)\sin^2(\theta)d\varphi^2\right],$$

from which we get the necessary quantities as

$$\sqrt{-g} = \ell^4 \sin^{-4}(T)\sin^2(\psi)\sin(\theta).$$

Plugging them into equation (39), we get

$$\Bigg[-\frac{\ell^2\sin^2(\psi)\sin(\theta)}{\sqrt{-g}}\partial_T \sin^{-2}(T)\partial_T + \frac{\ell^2\sin^{-2}(T)\sin(\theta)}{\sqrt{-g}}\partial_\psi \sin^2(\psi)\partial_\psi + \\ + \frac{\ell^2\sin^{-2}(T)}{\sqrt{-g}}\partial_\theta \sin(\theta)\partial_\theta + \frac{\ell^2\sin^{-2}(T)}{\sqrt{-g}\sin(\theta)}\partial_\varphi^2\Bigg]\Phi = m^2\Phi.$$

We divide through by $\frac{\ell^2 \sin^2(\psi)\sin(\theta)}{\sqrt{-g}}$, to get

$$\left[-\partial_T \sin^{-2}(T)\partial_T + \frac{\sin^{-2}(T)}{\sin^2(\psi)}\partial_\psi \sin^2(\psi)\partial_\psi + \frac{\sin^{-2}(T)}{\sin^2(\psi)\sin(\theta)}\partial_\theta \sin(\theta)\partial_\theta + \right.$$
$$\left. + \frac{\sin^{-2}(T)}{\sin^2(\psi)\sin^2(\theta)}\partial_\varphi^2 \right] \Phi = \sqrt{-g}\frac{\sin^{-2}(\psi)}{\sin(\theta)} m^2 \ell^{-2}\Phi.$$

A slightly more insightful and developed form for this equation is

$$\left[-\partial_T \sin^{-2}(T)\partial_T + \sin^{-2}(T) \left(\frac{1}{\sin^2(\psi)}\partial_\psi \sin^2(\psi)\partial_\psi + \right.\right.$$
$$\left.\left. + \frac{1}{\sin^2(\psi)\sin(\theta)}\partial_\theta \sin(\theta)\partial_\theta + \frac{1}{\sin^2(\psi)\sin^2(\theta)}\partial_\varphi^2 \right) \right] \Phi = \sin^{-4}(T) m^2 \ell^{-2}\Phi,$$

where we can now use

$$\frac{1}{\sin^2(\psi)}\partial_\psi \sin^2(\psi)\partial_\psi + \frac{1}{\sin^2(\psi)\sin(\theta)}\partial_\theta \sin(\theta)\partial_\theta + \frac{1}{\sin^2(\psi)\sin^2(\theta)}{\partial_\varphi}^2 = \nabla^2_{S_3}$$

to get the final form of our equation, namely

$$\left[-\partial_T \sin^{-2}(T)\partial_T + \sin^{-2}(T)\nabla^2_{S_3} \right] \Phi = \sin^{-4}(T) m^2 \ell^2 \Phi\,.$$

References

[1] “De Sitter Musings”, Dionysios Anninos, arXiv:1205.3855v2 (2012)

[2] “Conserved Charges in Asymptotically de Sitter Spacetimes”, P.B Aneesh, J. K Jahanur Hoque, A. Virmani, arXiv:1902.07415v3 (2019)

[3] “De Sitter Lectures”, Thomas Hartman, Cornell University

[4] “Conservation of Energy in de Sitter”, Ramsay Duff, Thesis (2021)

[5] “Classical Geometry of de Sitter Spacetime : an Introductory Review”, Y. Kim, C. Y. Oh, N.Park, arXiv:hep-th/0212326 (2002)

[6] “Conserved Quantities in de Sitter Spacetime”, S Jäger, Thesis (2008)

[7] “The Conformal Killing Equation on Forms - Prolongations and Applications”, A. Rod Gover, J. Šilhan, arXiv:math/0601751 (2008)

[8] “Vacuum States in de Sitter Space”, B. Allen, 10.1103/PhysRevD.32.3136 (1985)

[9] “Quantum Fields in Curved Space”, N. D. Birrel, P. C. W. Davies, Cambdriged Monographs on Mathematical Physics (1982)

[10] “Global Aspects of the Cauchy Problem, Y. Choquet-Bruhat, R. Geroch, Commun. math. Phys. 14, 329–335 (I969)

[11] “Structure and Geometry of Lie Groups”, J. Hilgert, K. Neeb, Springer Monographs in Mathematics (2012)

[12] “Wave Function of the Universe”, J. B. Hartle, S. W. Hawking, Phys. Rev. D 28, 2960 (1983)

[13] “Recurrent Nightmares? : measurement Theory in de Sitter Space”, T. Banks, W. Fischler, S. Paban, arXiv:hep-th/0210160v2 (2002)

[14] “ Nonuniqueness of Canonical Field Quantization in Riemannian Space-Time” , Stephen A. Fulling Phys. Rev. D 7, 2850 (1973)

[15] “Quantum fields in curved spacetime”, S. Holland, R. Wald, arXiv:1401.2026v2 (2014)

[16] “Quantum Field Theory in De Sitter Space: Renormalization by Point-Splitting”, T. S. Bunch, P. C. W. Davies, Proceedings of the Royal Society of London. Series A, Mathematical and Physical Sciences Vol. 360, No. 1700 (1978)

[17] “Fun and Supersymmetry”, Flip Tanedo, Institute for High Energy Phenomenology (2010)

[18] “Hankel functions. Encyclopedia of Mathematics.”
URL: http://encyclopediaofmath.org/index.php?title=Hankel_functions&oldid=31410

[19] “Laplacian Eigenmodes for the Three Sphere”, M. Lachièze-Rey, arXiv:math/0401153 (2004)

[20] “Consequences of Field Quantization in De Sitter Type Cosmological Models”, E.A. Tagirov, Ann. Phys., 76 (1973)

[21] “Cosmological event horizons, thermodynamics and particle creation”, G.W. Gibbons, S.W. Hawking, Phys. Rev. D 15, 2738 (1977)

[22] “Decoupling in an expanding universe : boundary RG-flow affects initial conditions for inflation”, K. Schalm, G. Shiu, J. P. van der Schaar, Journal of High Energy Physics, Volume 2004, JHEP04 (2004)

[23] “Towards the non-perturbative cosmological bootstrap”, M. Hogervorst, J. Penedones, K. S. Vaziri, arXiv:2107.13871v1 (2021)

[24] “ Notes on $\tilde{\mathrm{SL}}(2,R)$ representations”, A. Kitaev, arXiv:1711.08169v2 (2018)

[25] “Conformal Vacua and Entropy in de Sitter Space”, R. Bousso, A. Maloney, A. Strominger, arXiv:hep-th/0112218v1 (2001)

[26] “Aspects of Asymptotically de Sitter Spacetimes Lectures”, Dionysios Anninos, Unpublished

[27] “Massless minimally coupled scalar field in de Sitter space”, B. Allen, A. Folacci, Phys. Rev. D 35, 3771 (1987)

[28]“TASI Lectures on Inflation”, Daniel Baumann, arXiv:0907.5424v2 (2012)

[29] “The IR stability of de Sitter: Loop corrections to scalar propagators”, D. Marolf, I. A. Morrison, arXiv:1006.0035v3 (2010)

[30] “De Sitter Breaking through Infrared Divergences”, S. P. Miao, N. C. Tsamis, R. P. Woodard, arXiv:1002.4037v1 (2010)

[31] “Massless scalar field vacuum in de Sitter spacetime”, D. N. Page, X. Wu, arXiv:1204.4462v5 (2012)

[32] “Quantization of the massless minimally coupled scalar field and the dS/CFT correspondence”, A. J. Tolley , N. Turok, arXiv:hep-th/0108119v2 (2002)

[33] “Cluster Decomposition Properties of the S Matrix”, E. H. Wichmann, J. H. Crichton, Phys. Rev. 132, 2788 (1963)

[34] “Future and Origin of our Universe: Modern View”, A.A. Starobinsky, arXiv:astro-ph/9912054v1 (1999)

[35] “Cosmic clustering”, D. Anninos, F. Denef, J. High Energ. Phys. 2016, 181 (2016)

[36] “CMB Temperature and Polarization Anisotropy Fundamentals”, W. Hu, arXiv:astro-ph/0210696v1 (2002)

[37] “Cosmic Microwave Background Anisotropies”, W. Hu, S. Dodelson, arXiv:astro-ph/0110414v1 (2001)

[38] “The Function of Boundary Conditions in Physical Sciences”, J.R. S. Bursten, Philosophy of Science, 88(2), 234-257 (2021)

[39] “Future Evolution of Structure in an Accelerating Universe”, M. T. Busha, F. C. Adams, R. H. Wechsler, A. E. Evrard, arXiv:astro-ph/0305211v1 (2003)

[40] “Status of Cold Dark Matter Cosmology”, J. R. Primack, arXiv:astro-ph/0205391v1 (2002)

[41] “Multiple Peaks in the Angular Power Spectrum of the Cosmic Microwave Background: Significance and Consequences for Cosmology”, P. de Bernardis et al, ApJ 564 559 (2002)

[42] “The evolution of CMB spectral distortions in the early Universe”, J. Chluba, R. A. Sunyaev, Mon. Not. R. Astron. Soc. 419, 1294–1314 ((2012)

[43] “de Sitter Holography with a finite number of states”, M. Parikh, E. Verlinde, JHEP01 054 (2005)

[44] “The trouble with de Sitter space”, N. Goheer, M. Kleban, L. Susskind, JHEP07 056 (2003)

[45] “Massive holographic QFTs in de Sitter”, J. M. Penin , K. Skenderis, B. Withers, arXiv:2112.14639v2 (2022)

[46] “The dS/CFT Correspondence”, A. Strominger, arXiv:hep-th/0106113v2 (2001)

[47] “Higher Spin Realization of the dS/CFT Correspondence”, D. Anninos, T. Hartman, A. Strominger, arXiv:1108.5735v1 (2011)

[48] “Obstacles to Constructing de Sitter Space in String Theory”, Michael Dine, J. A.P. Law-Smith, S. Sun, D. Wood, Y. Yu, arXiv:2008.12399v4 (2021)

[49] “de Sitter vacua in string theory”,S. Kachru, R. Kallosh, A. Linde, S. P. Trivedi, Phys. Rev. D 68, 046005 (2003)

[50] “Derivation of the Wheeler-Dewitt equation from a path integral for minisuperspace models”, J. J. Halliwell, Phys. Rev. D 38, 2468 (1988)

[51] “Late-time structure of the Bunch-Davies de Sitter wavefunction”, D. Anninos et al, JCAP11 048 (2015)

[52] “Infrared problem in the Faddeev-Popov ghost propagator in perturbative quantum gravity in de Sitter spacetime”, J. Gibbons, A. Higuchi , W. C. C. Lima

[53] “The Cosmological Constant Problem and Running Vacuum in the Expanding Universe”, J. S. Peracaula, arXiv:2203.13757v1 (2022)